

BIOLOGY REVIEW

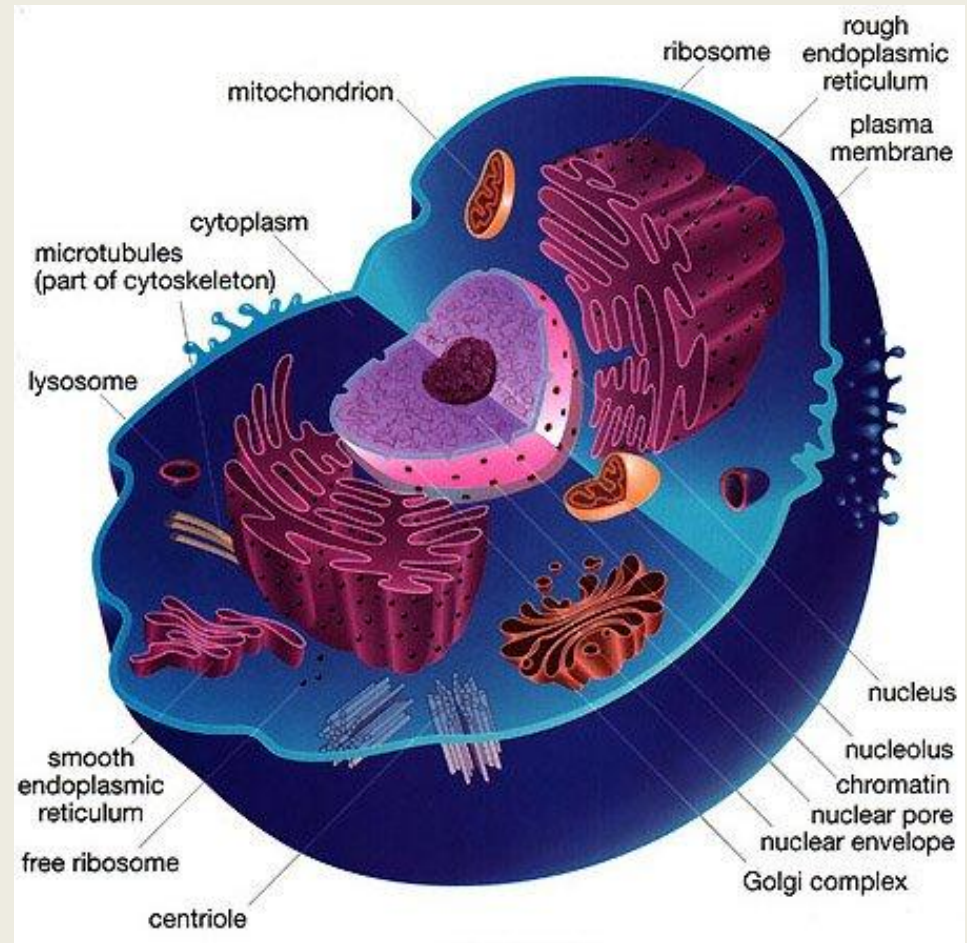
UPCAT 2012

CONTENTS

- The Cell
- Genetics
- The Chemistry of Life
- Evolution and Ecology
- Form and Function

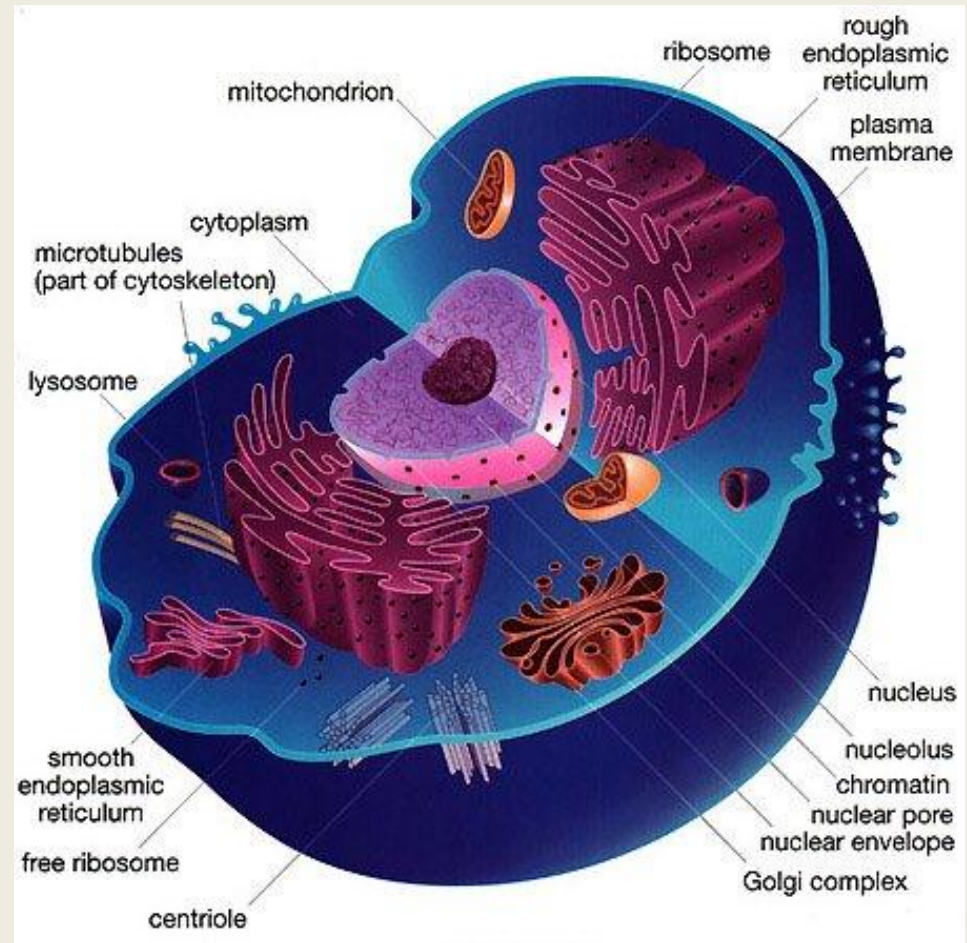
THE CELL

- Found in great numbers in an active cell
- Contains genetic information that directs activities of the cell
- Network of membrane-enclosed spaces involved in protein manufacture



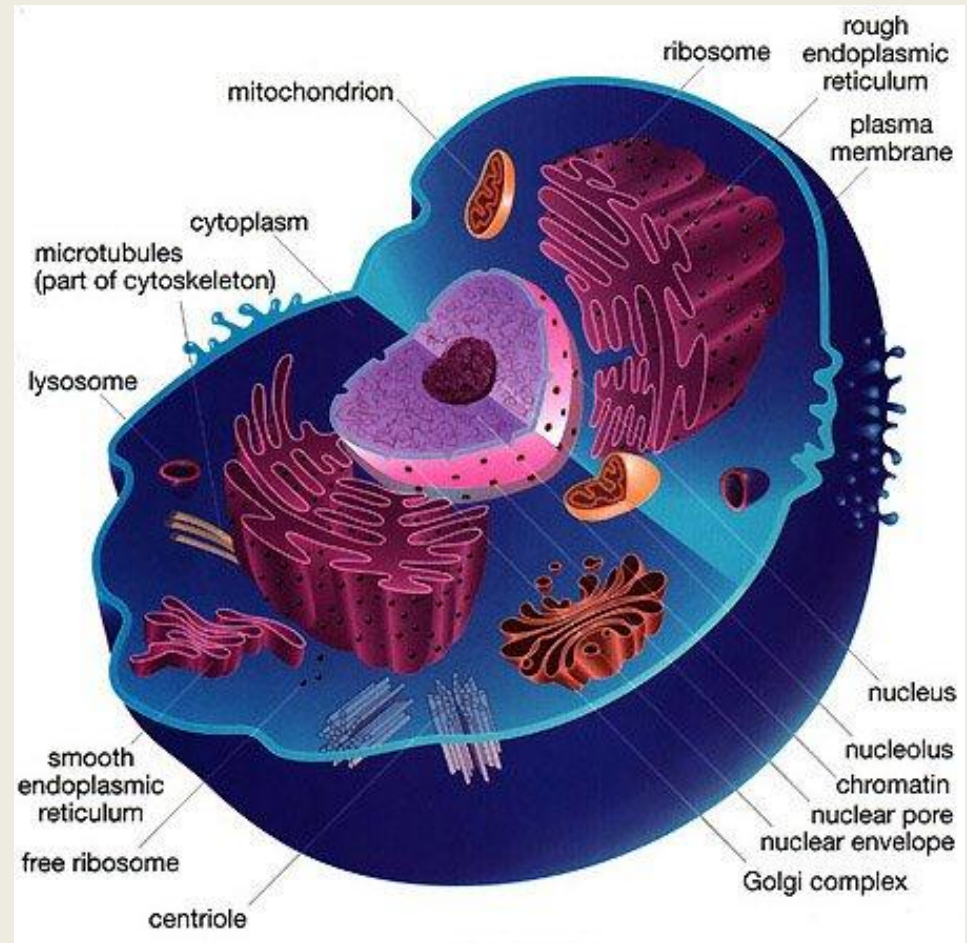
THE CELL

- Freely-floating structures in cytoplasm involved in protein synthesis
- Phagocytic cells have this in abundance
- Controls transfer of substances into and out of cells



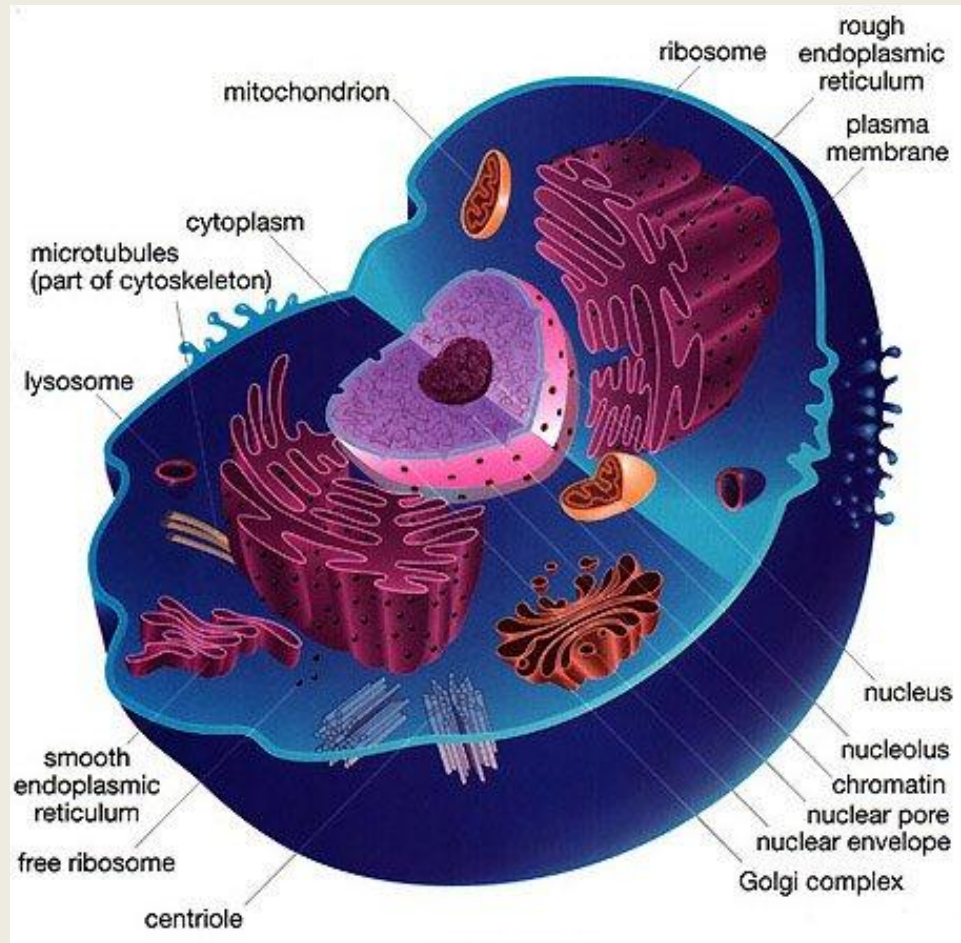
THE CELL

- Site of lipid formation
- Reinforces cell's shape and functions in cell movement
- Active in protein modification, sorting, and secretion of cell products



THE CELL

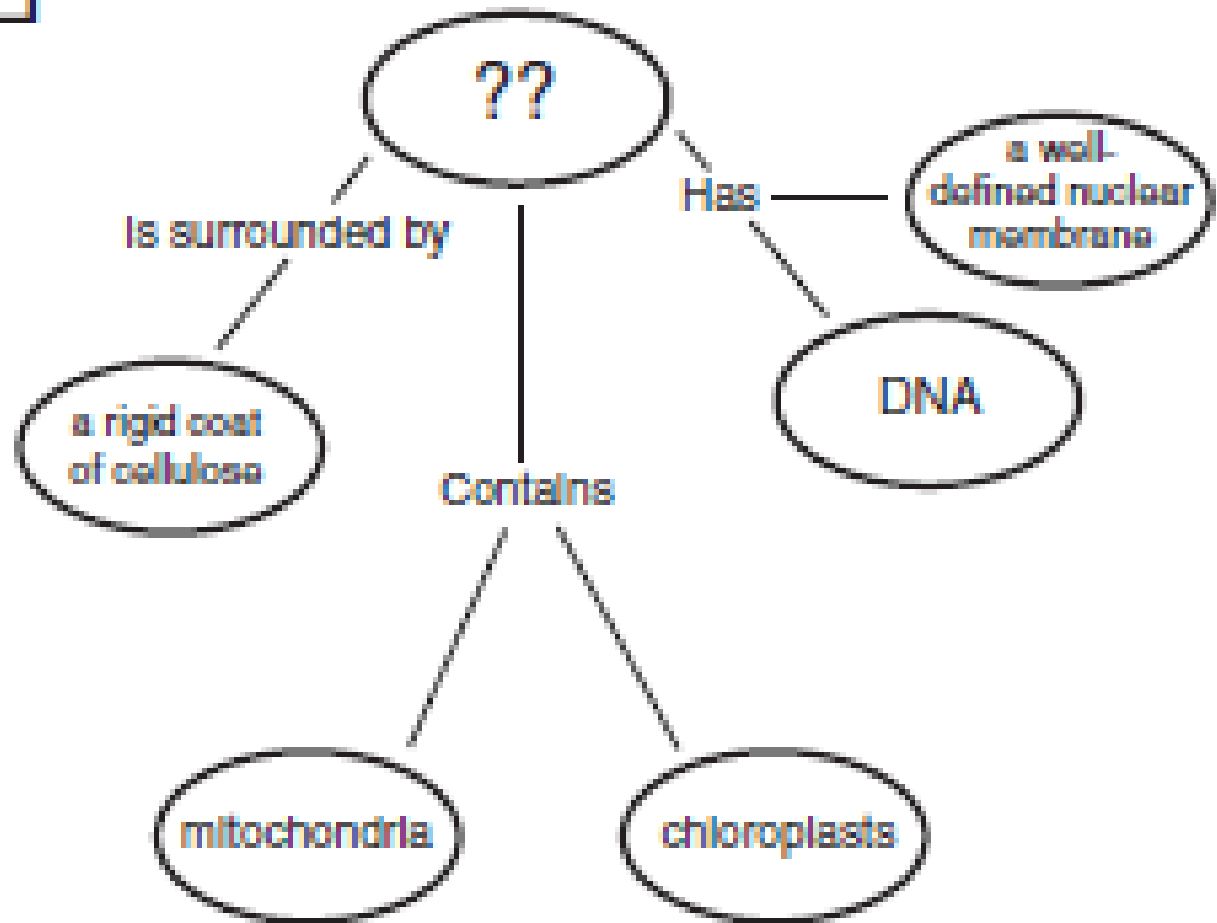
- Involved in production of ribosomes
- Consists of DNA and proteins



THE CELL

16

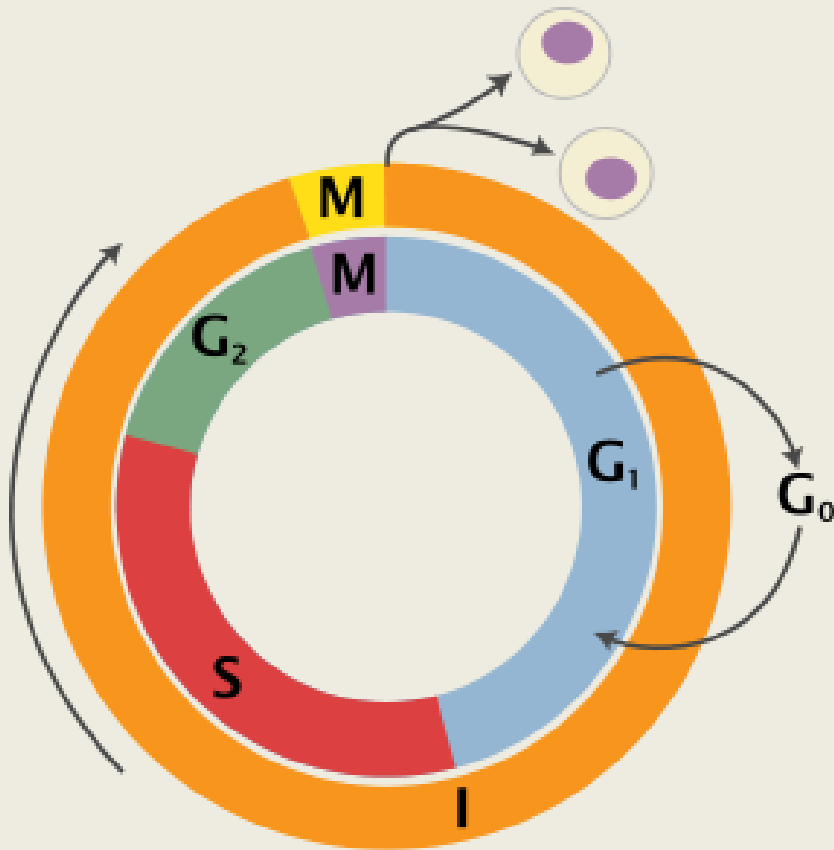
- What type of cell is described by this concept map?



THE CELL

12.	Prokaryote	Eukaryote
Cell wall		
Cell membrane		
Cytoplasm		
Lysosomes		
Centrosomes/centrioles		
Flagella		
Chloroplasts		
Membrane-bound nucleus		
Vacuole		
Size		

CELL CYCLE

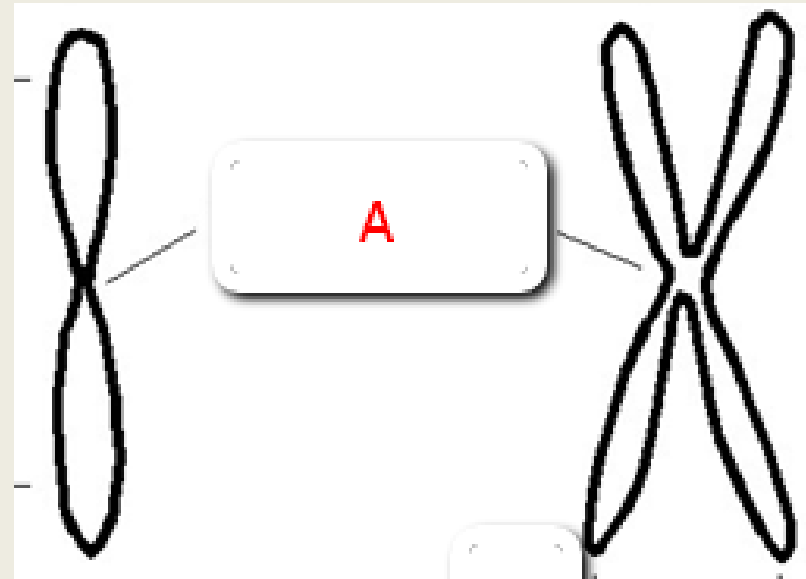


In which part of the cycle does:

- Cell increase in size
- Each daughter cell receives its own copy of the parent cell's DNA
- DNA replicate
- The cell leave the cycle and stop dividing
- The cell produce microtubules required for division

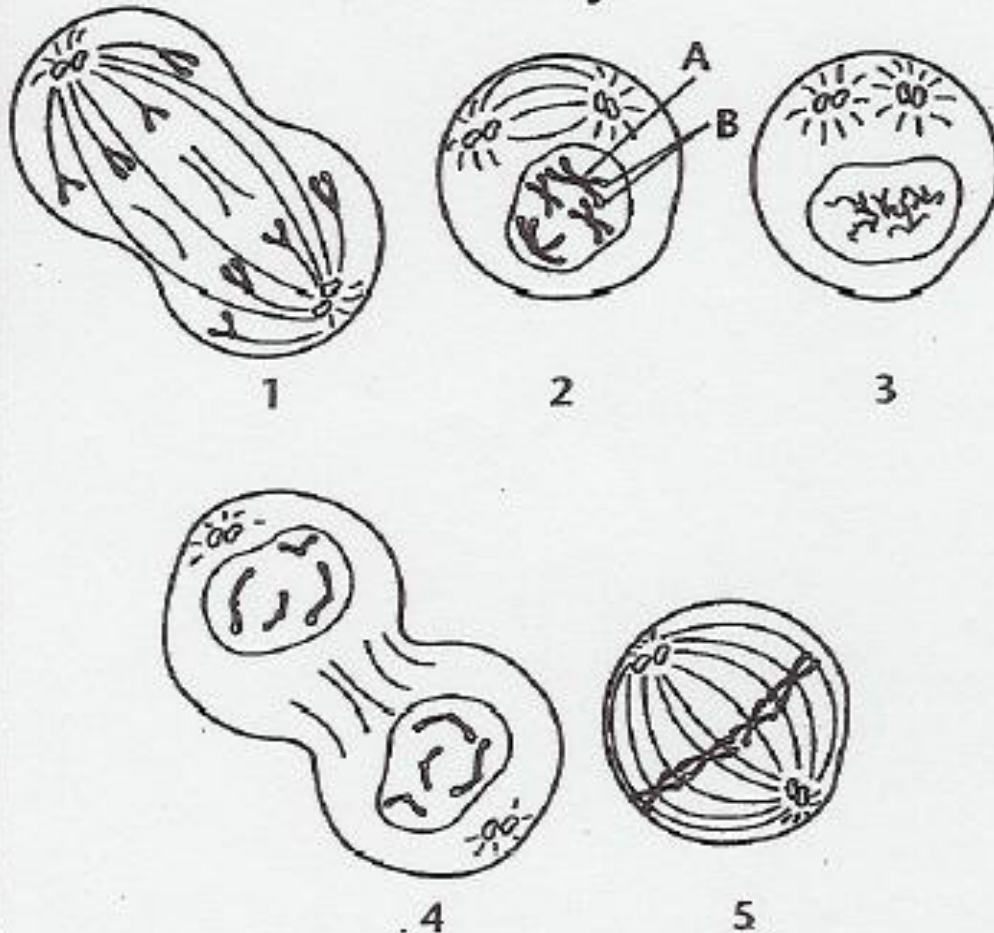
CELL CYCLE

- What is the difference between the two chromosomes shown?
- Locate the following:
 - Centromere
 - Sister chromatid
 - Gene



CELL DIVISION

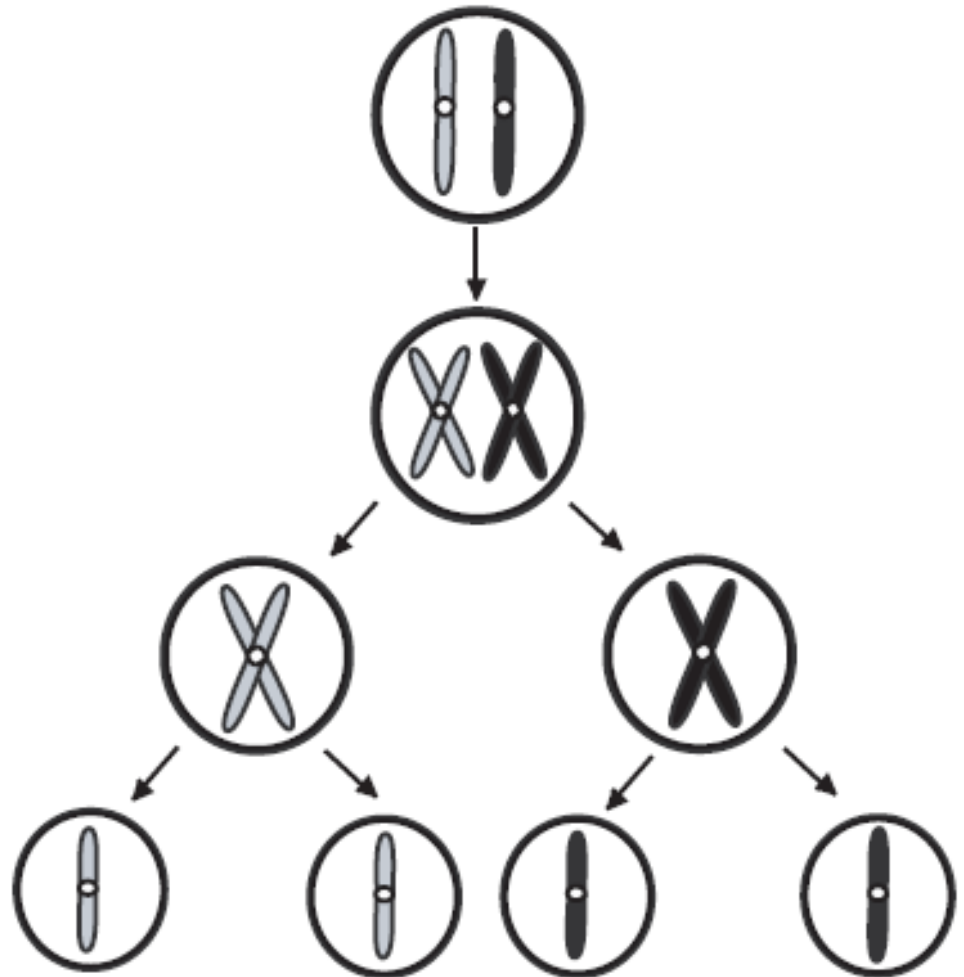
The Cell Cycle



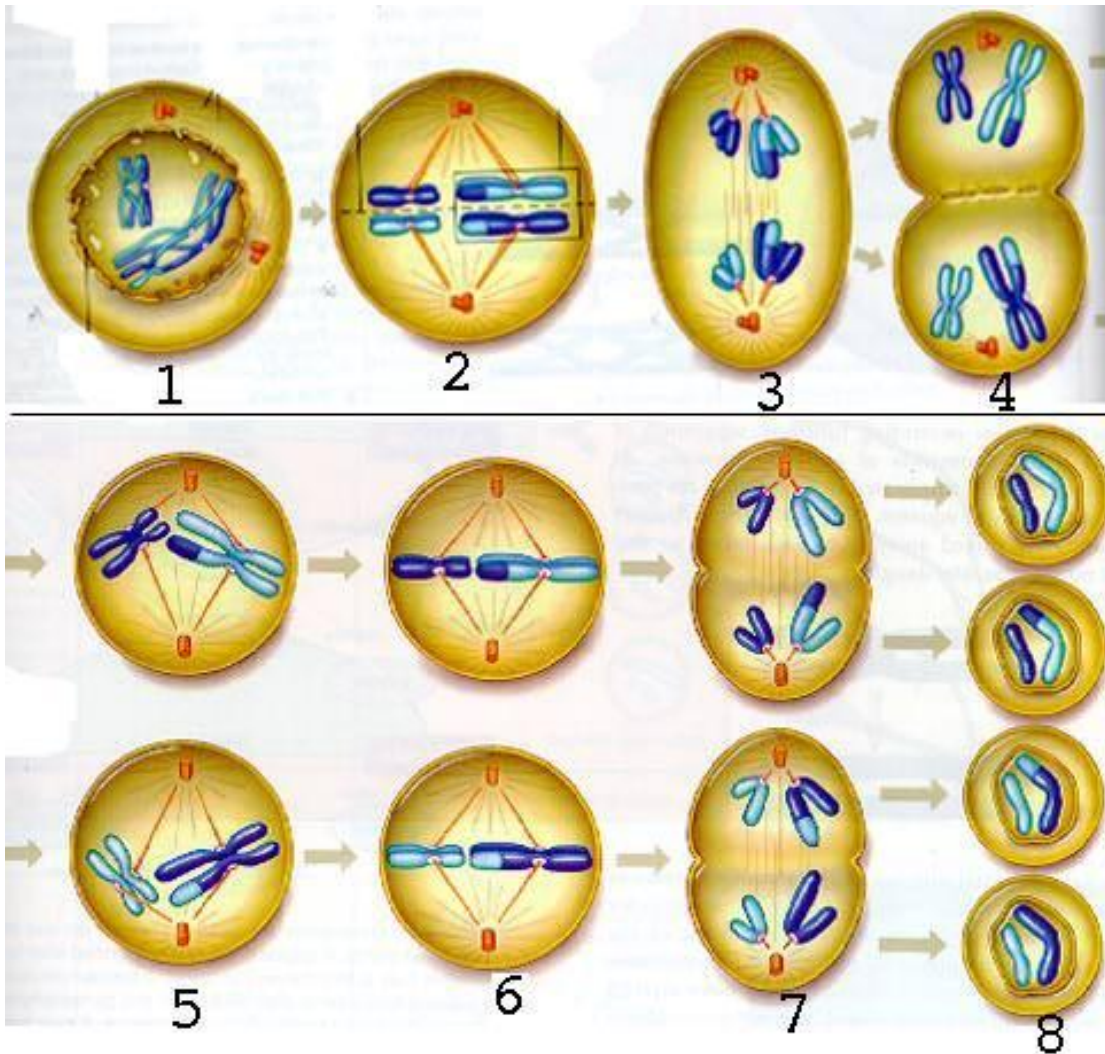
- Sort the diagrams according to chronological sequence.
- Identify the stages shown in each section.
- What process is shown here?
- How many chromosomes and chromatids are present at each stage?

CELL DIVISION

- What process is shown on the right?
- What is the ploidy level of the parent cell?
- What is the ploidy level of the daughter cells?
- Where does this process occur?

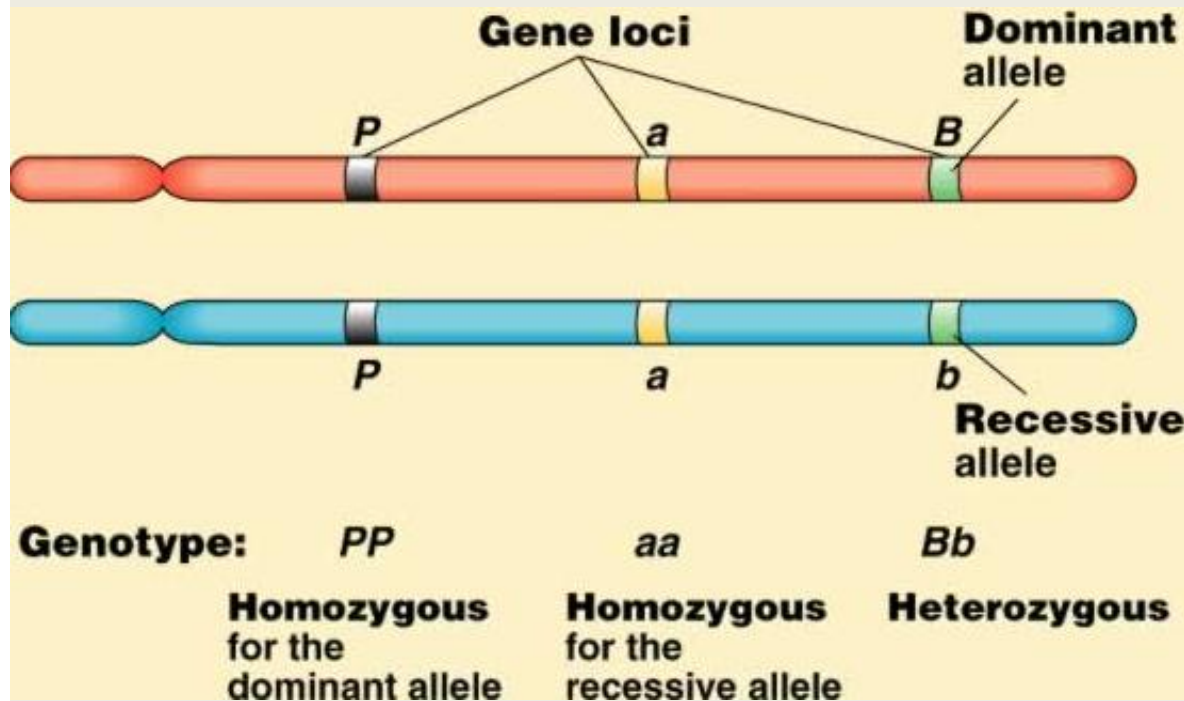


CELL DIVISION



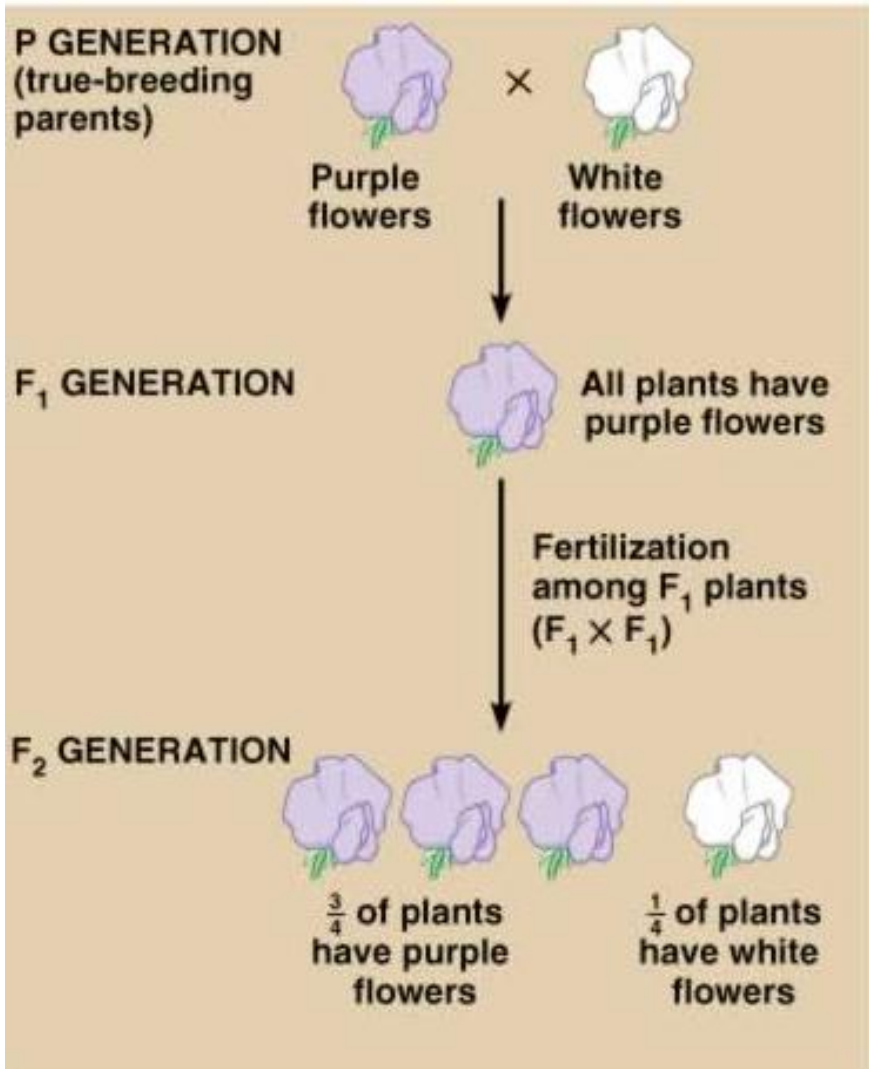
- Identify the stages of the process.
- Differentiate between meiosis I and mitosis.
- Differentiate between meiosis II and mitosis.
- What happens to the four daughter cells in a male? In a female?

GENETICS



- What relationship does the blue chromosome have with the red chromosome?
- What is the difference between a locus and an allele?
- What is the difference between a dominant and a recessive allele?
- How are dominant and recessive alleles symbolized?
- What is the difference between genotype and phenotype?

GENETICS



- Show what is happening using conventional genetic symbols.
- In the F₂ generation, what percentage of the offspring are true-breeding plants?

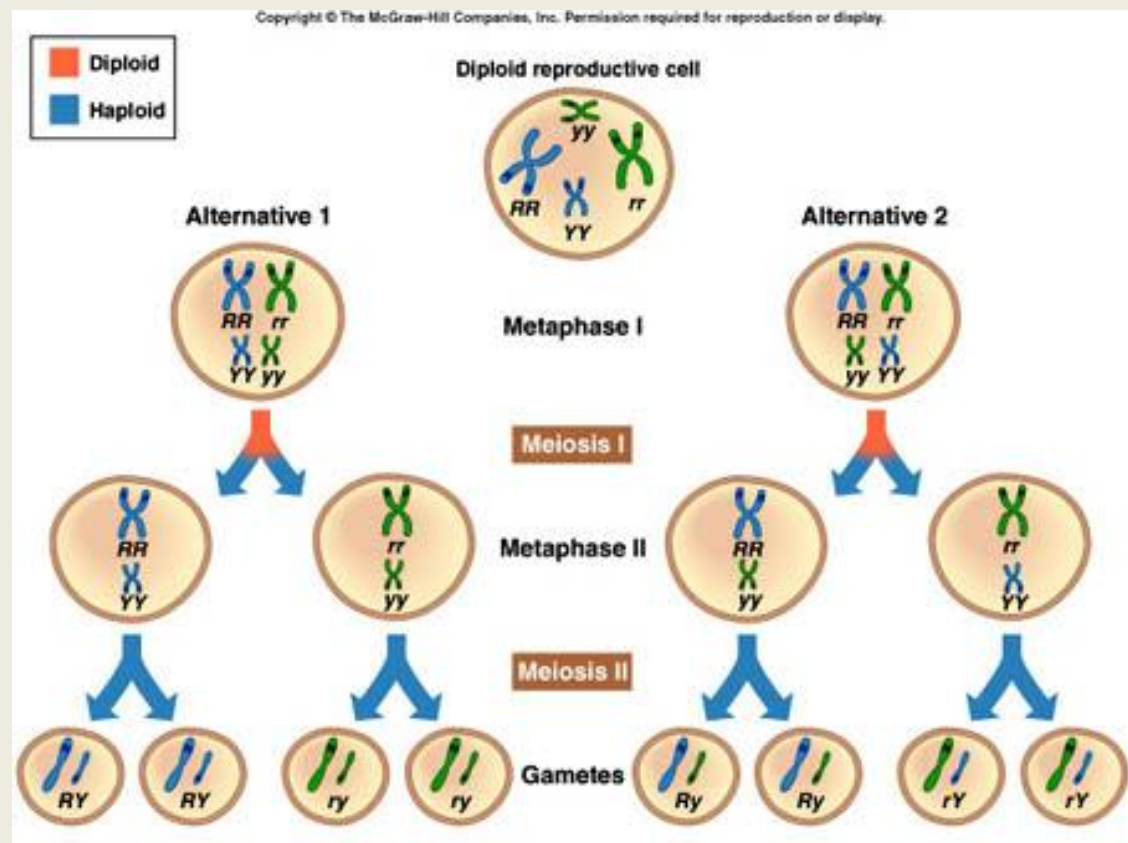
GENETICS

In peas, yellow seed color is dominant to green.

- If a true-breeding yellow pea plant is crossed with a true-breeding green pea plant, what are the possible phenotypes/genotypes of the offspring?
- If two members of the F₁ generation from the cross above are crossed with each other,
 - what are the chances of obtaining offspring that are identical to the green parent?
 - what are the chances of obtaining heterozygous offspring?

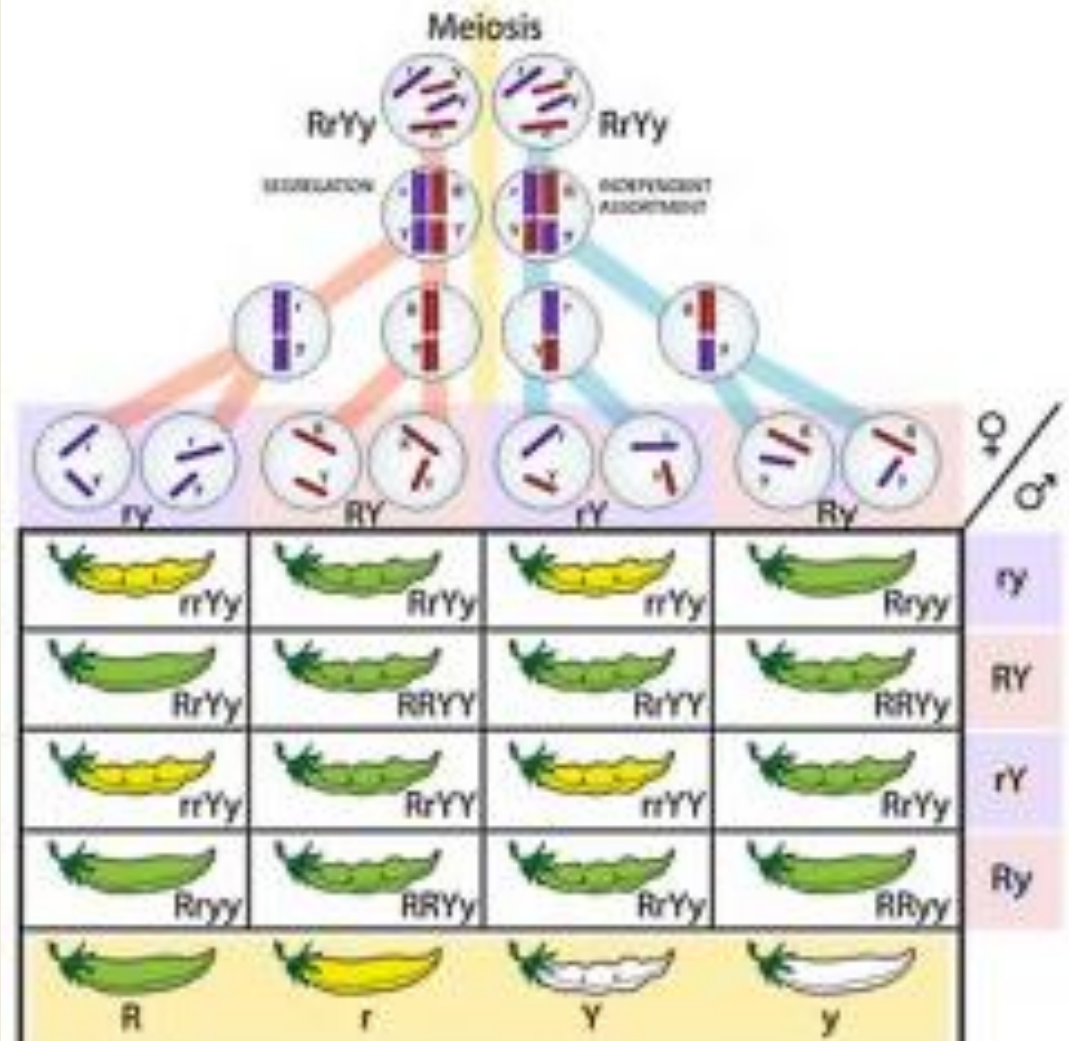
GENETICS

- **Mendel's Law of Segregation:** two alleles for each gene separate during gamete formation
- **Mendel's Law of Independent Assortment:** separate genes for separate traits are passed independently of one another



GENETICS

- **Mendel's Law of Segregation:** two alleles for each gene separate during gamete formation
- **Mendel's Law of Independent Assortment:** separate genes for separate traits are passed independently of one another



GENETICS

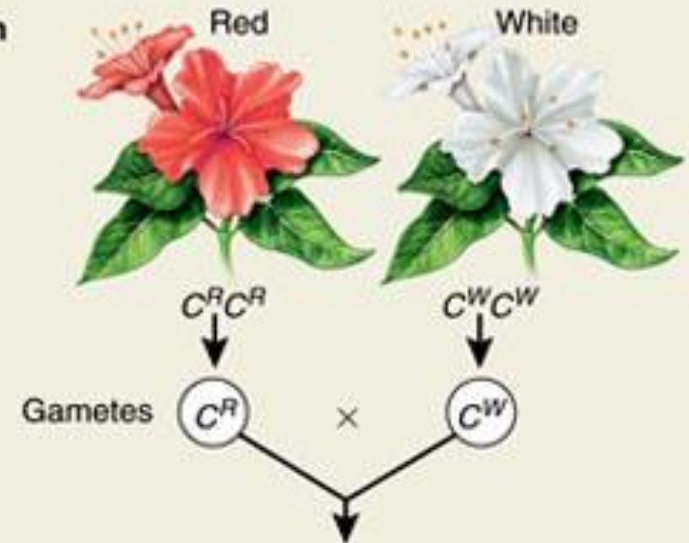
In peas, yellow seed color is dominant to green and the round seed shape is dominant to the wrinkled seed shape.

- If a true-breeding yellow, wrinkled pea plant is crossed with a true-breeding green, round pea plant, what are the possible phenotypes/genotypes of the offspring?
- If two members of the F₁ generation from the cross above are crossed with each other,
 - what are the chances of obtaining offspring that appear yellow and round?
 - what are the chances of obtaining offspring that are phenotypically identical to the parental generation?

GENETICS

- What type of dominance is exhibited?
- What are the possible genotypes and phenotypes of the F_2 generation?

P generation

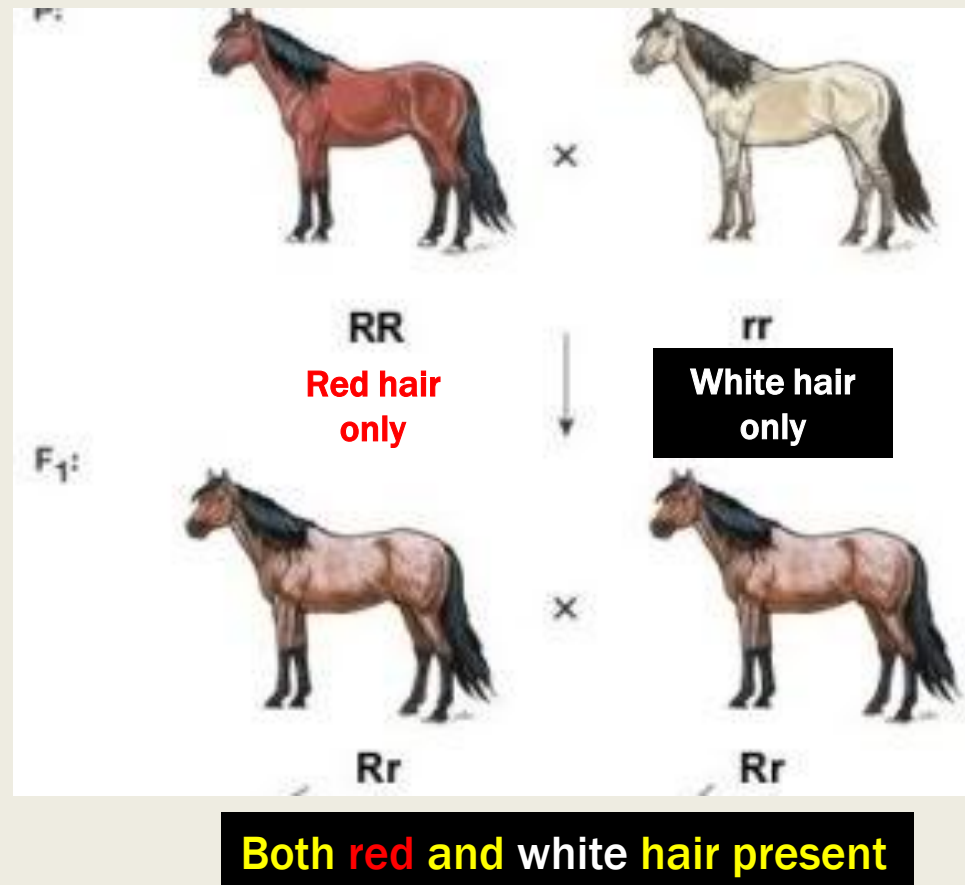


F_1 generation



GENETICS

- What type of dominance is exhibited?
- What are the possible genotypes and phenotypes of the F_2 generation?

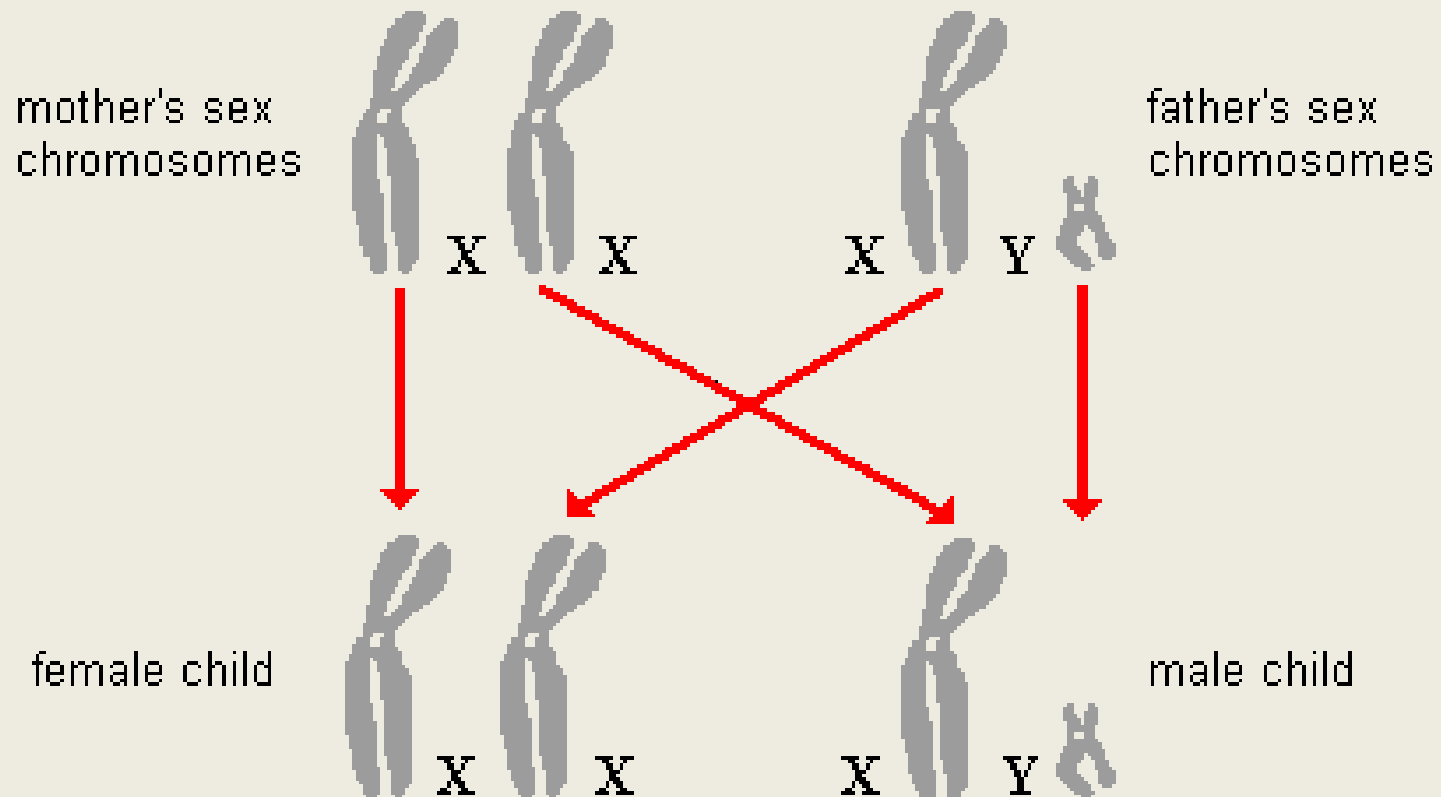


GENETICS

- What are the phenotypes and genotypes of offspring produced by a mating between
 - Two type O individuals
 - Two type AB individuals
 - A type A and a type O individual
- A mother with type A and a father with type B cannot have an offspring with type _____ blood.

Blood type of cells	Genotype
A	$I^A I^A$ or $I^A i^O$
B	$I^B I^B$ or $I^B i^O$
AB	$I^A I^B$
O	$i^O i^O$

GENETICS

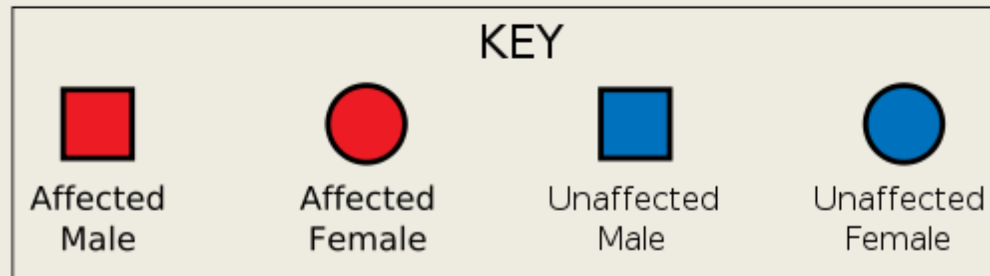
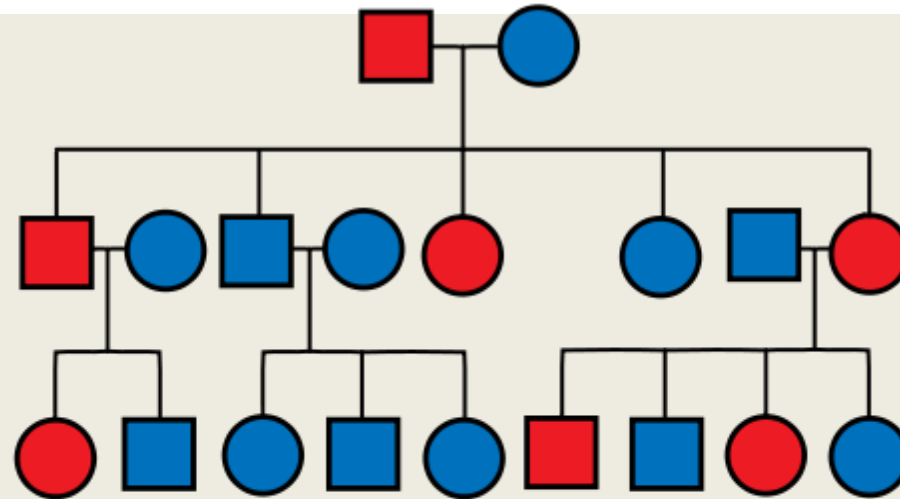


- What happens when the locus for a particular character is located on a sex chromosome?

GENETICS

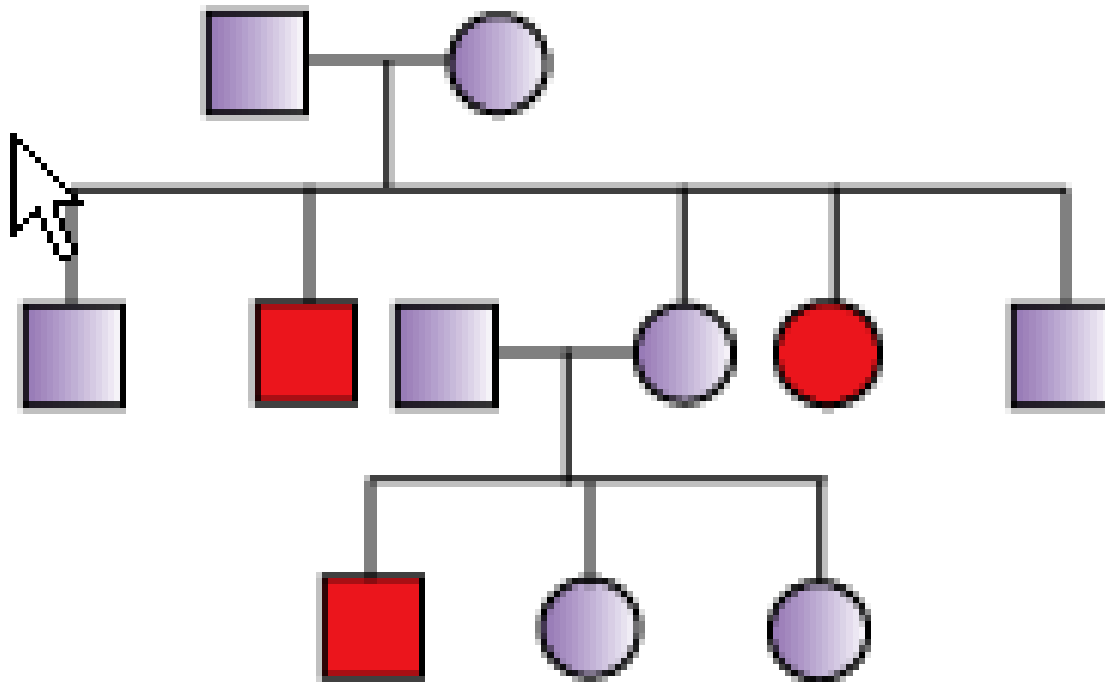
- In humans, pseudohypertrophic muscular dystrophy is a condition in which the muscles gradually waste away, ending in death in the early teens. In some families it is dependent upon a x-linked recessive gene. This type occurs only in boys and has never been reported in girls. Why is it not to be expected in girls?
- Hemophilia in humans is due to an X-chromosome mutation. What will be the results of mating between a normal female and a hemophiliac male?
- A human female "carrier" who is heterozygous for the recessive, sex-linked trait causing red-green color blindness, marries a normal male.
 - What proportion of their male progeny will have red-green color blindness?
 - What proportion of their female progeny will show the trait?

GENETICS



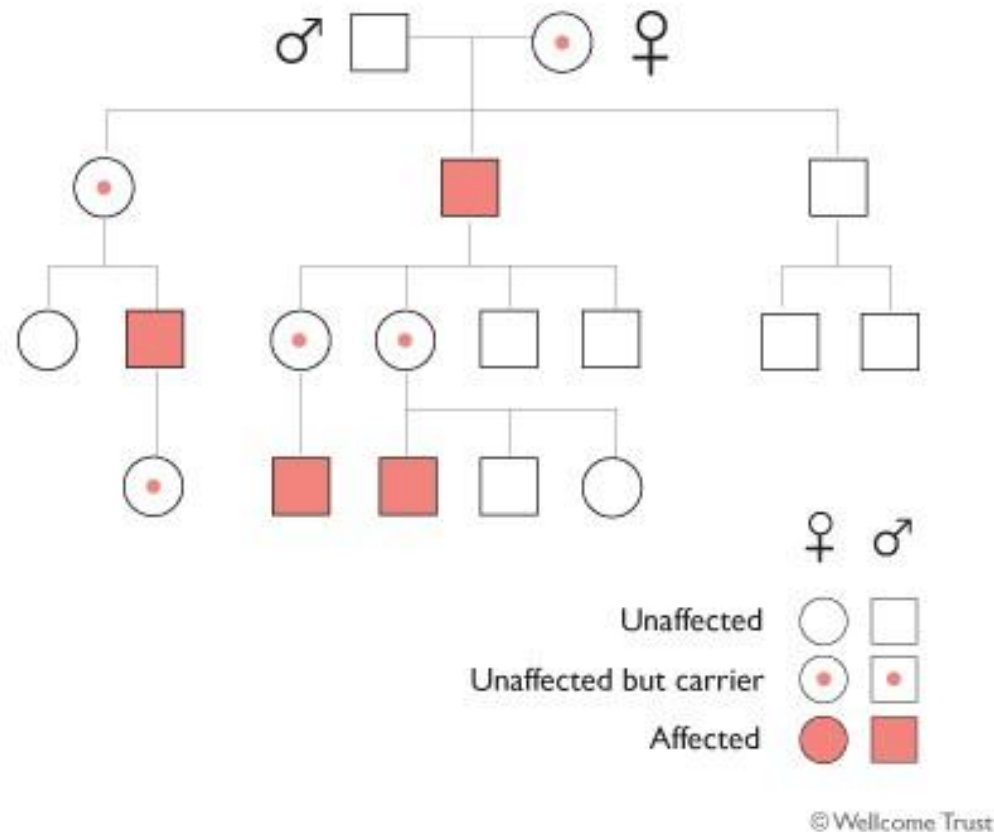
- What pattern of inheritance is shown by this pedigree chart?

GENETICS



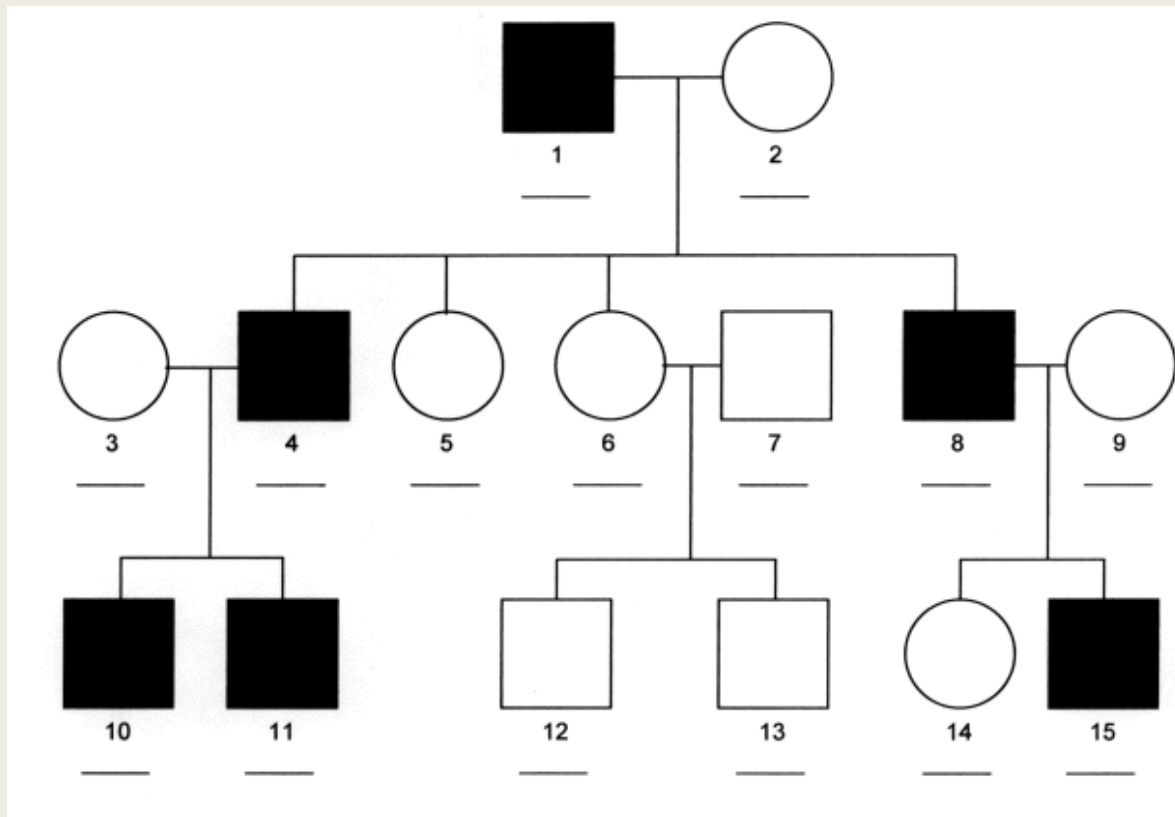
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GENETICS



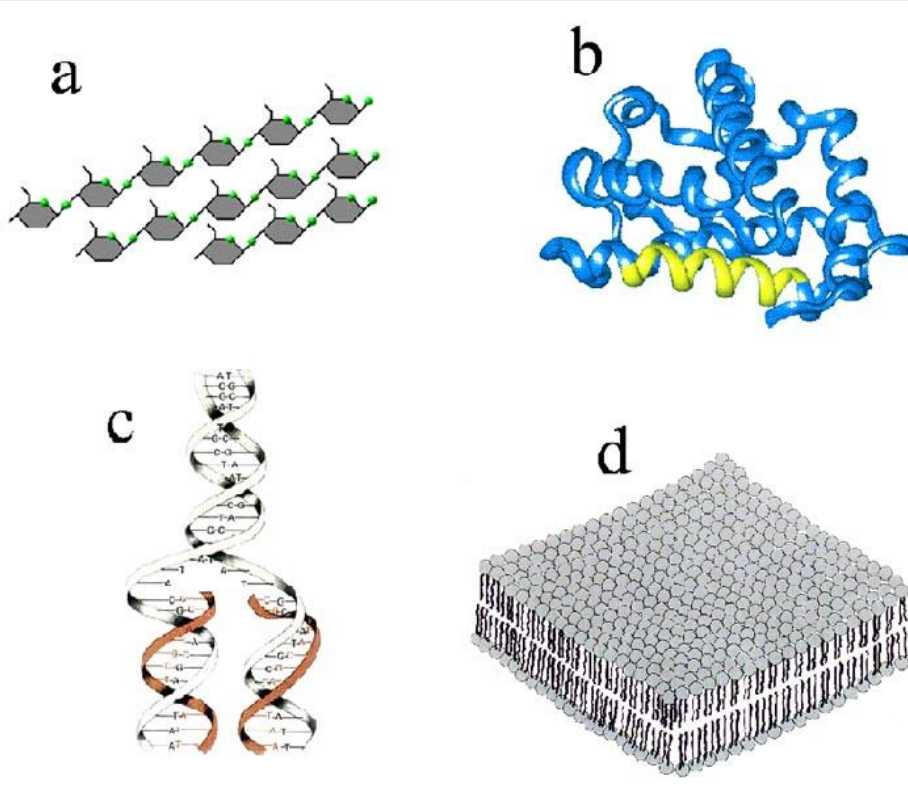
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GENETICS



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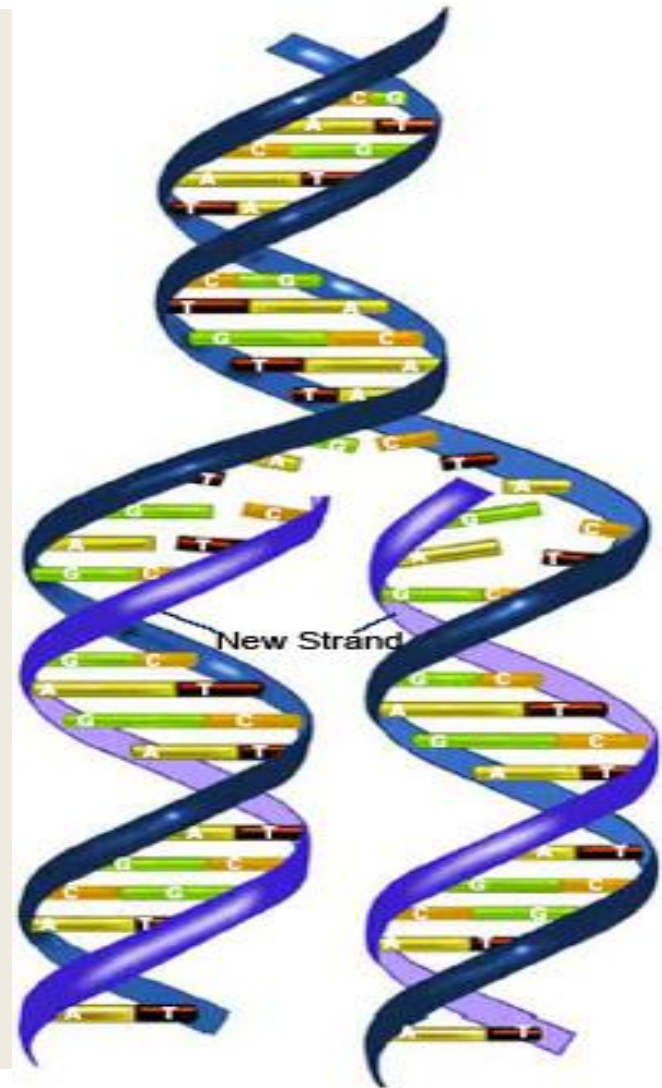
THE CHEMISTRY OF LIFE



- Not made up of repeating sub-units
- Form bilayers of membranes
- Chitin, cellulose, glucose, starch, and glycogen
- Enzymes, hormones; function for structure, movement, and transport
- Have many complex levels of structures (from primary to quaternary)
- Store, transmit, express hereditary information

THE CHEMISTRY OF LIFE

- A pure sample of plant DNA was analyzed and cytologists determined that adenine made up 23% of the entire sample. How much cytosine could be expected to be present in the same sample?



THE CHEMISTRY OF LIFE

		Second mRNA base				
		U	C	A	G	
First mRNA base (5' end)	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA Stop UAG Stop	UGU } Cys UGC } UGA Stop UGG } Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } CGC } Arg CGA } CGG }	U C A G
	A	AUU } AUC } Ile AUA } AUG Met or start	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

Third mRNA base (3' end)

DNA Sequence used as a template:

TAG TCT CCG TAC TCC CCT
GGA ACG CGA TCC ATT

Complementary Strand:

ATC AGA GGC ATG AGG GGA
CCT TGC GCT AGG TAA

RNA Strand:

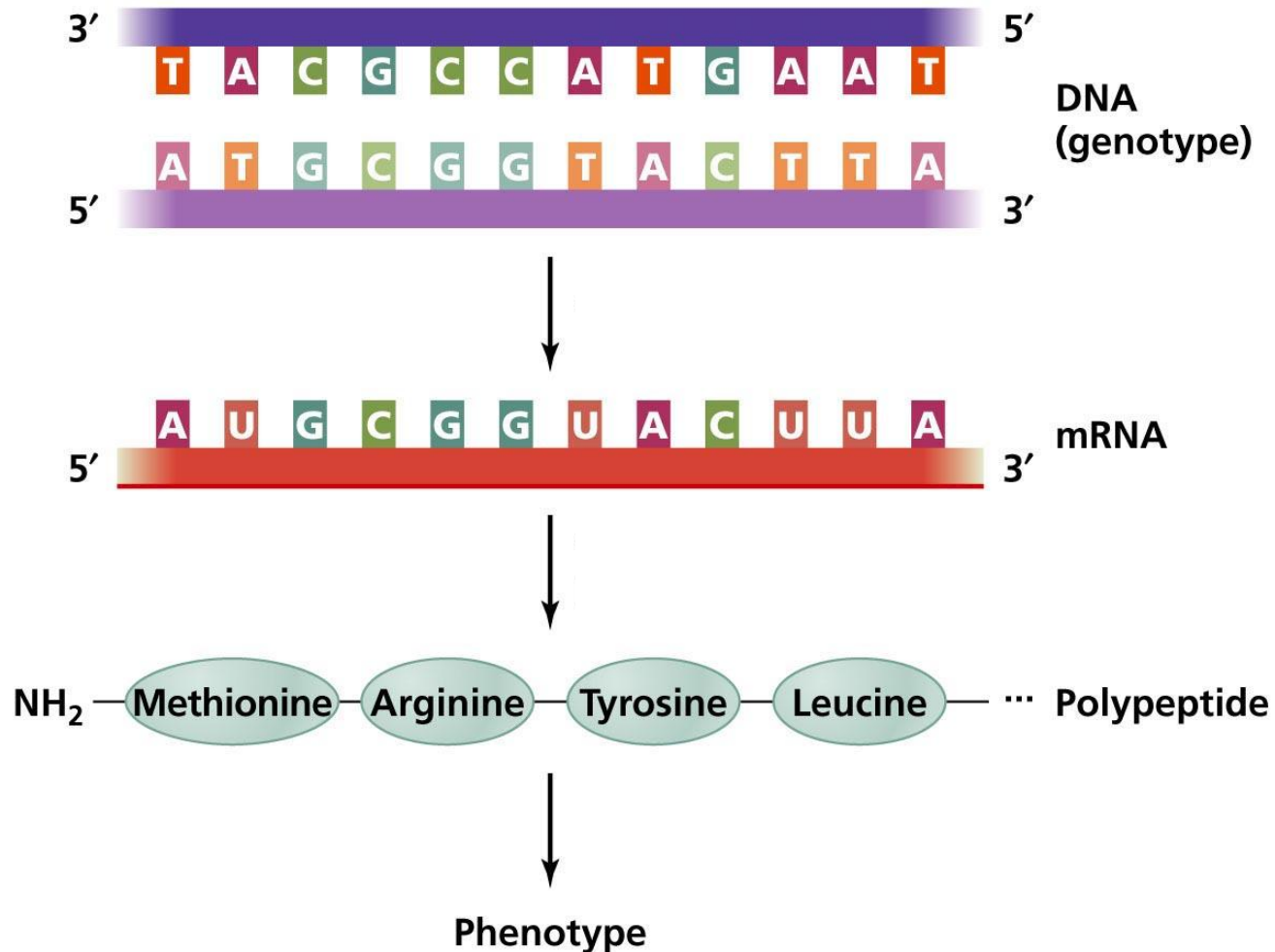
AUC AGA GGC AUG AGG GGA
CCU UGC GCU AGG UAA

Polypeptide Sequence

Met Arg Gly Pro Cys Ala
Arg STOP

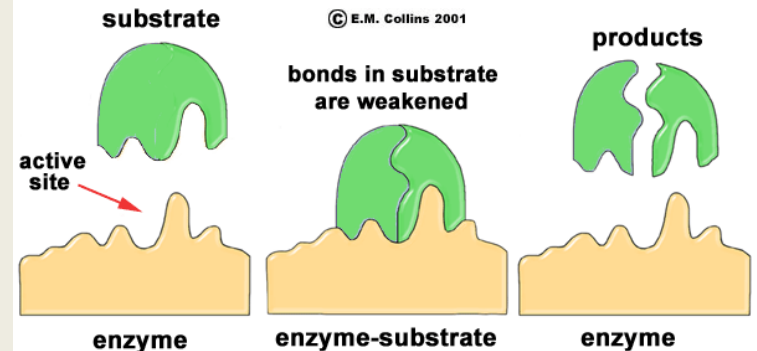
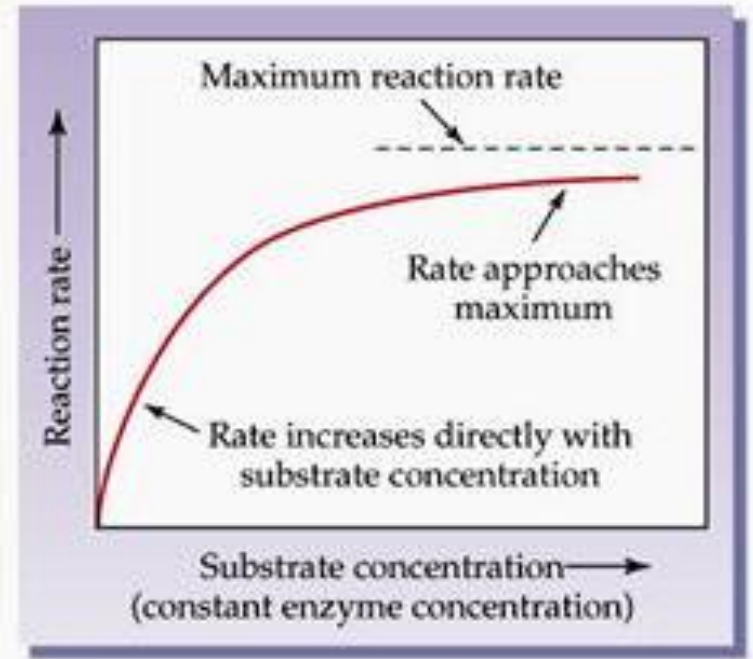
THE CHEMISTRY OF LIFE

- Semi-conservative process
- Occurs in the nucleus
- Occurs on ribosomes
- Composed of groups called codons



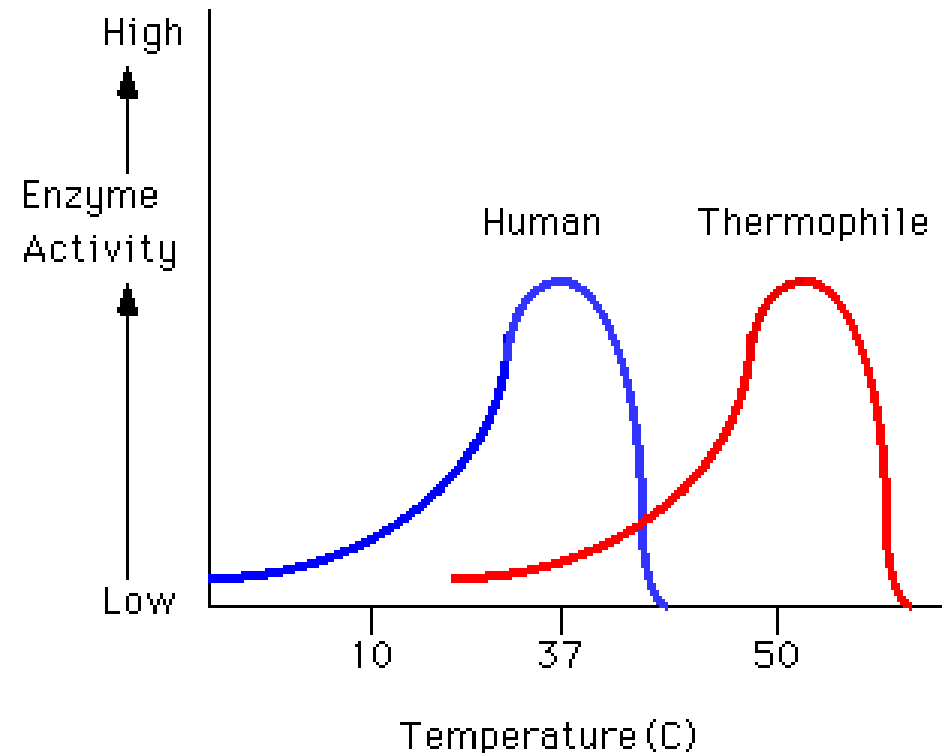
CHEMISTRY OF LIFE

- Salivary amylase is an enzyme that catalyzes the breakdown of starch into simple sugars. Starch was added in increasing amounts to a beaker with salivary amylase and the reaction rates were measured and plotted. What causes the graph to start leveling-off and how can the reaction rate be increased further?

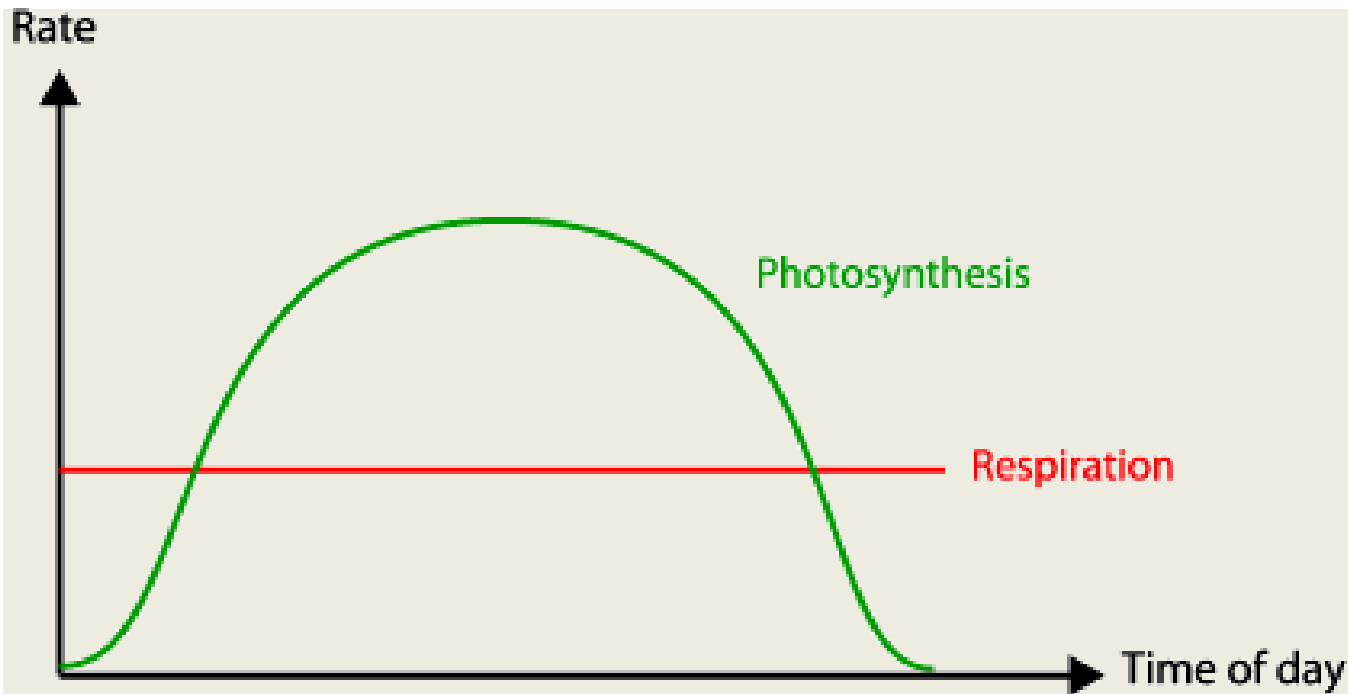


CHEMISTRY OF LIFE

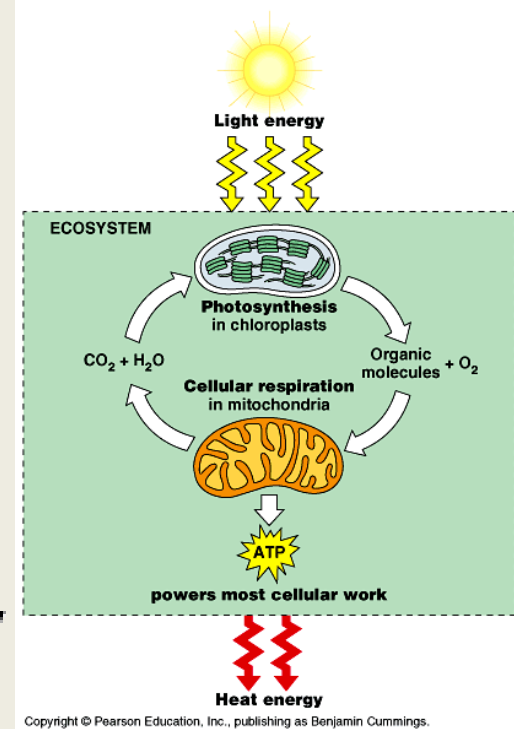
- Why do humans and thermophiles have graphs that fall within different ranges?
- What causes the characteristic shape of the graphs?



PHOTOSYNTHESIS AND RESPIRATION

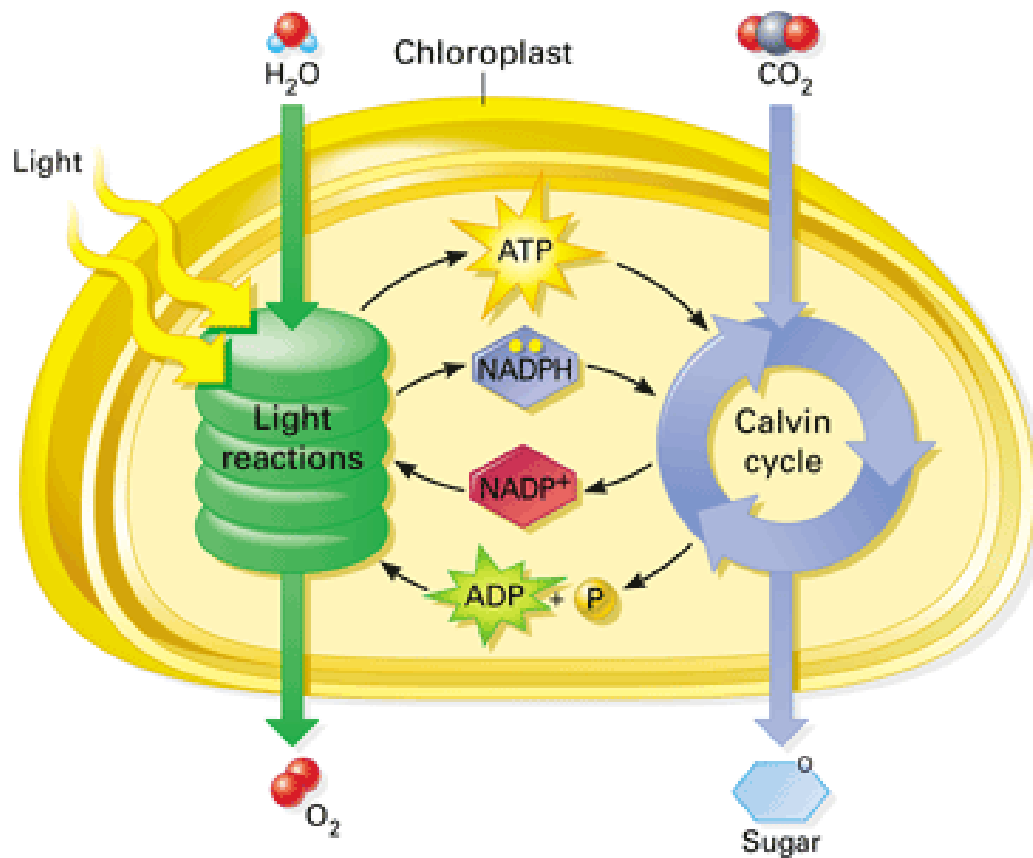


Graph one: the rates of photosynthesis & respiration over the daylight hours



- Explain why the graphs for photosynthesis and respiration rates have the following shapes.
- What units can be used to measure these rates?

PHOTOSYNTHESIS AND RESPIRATION

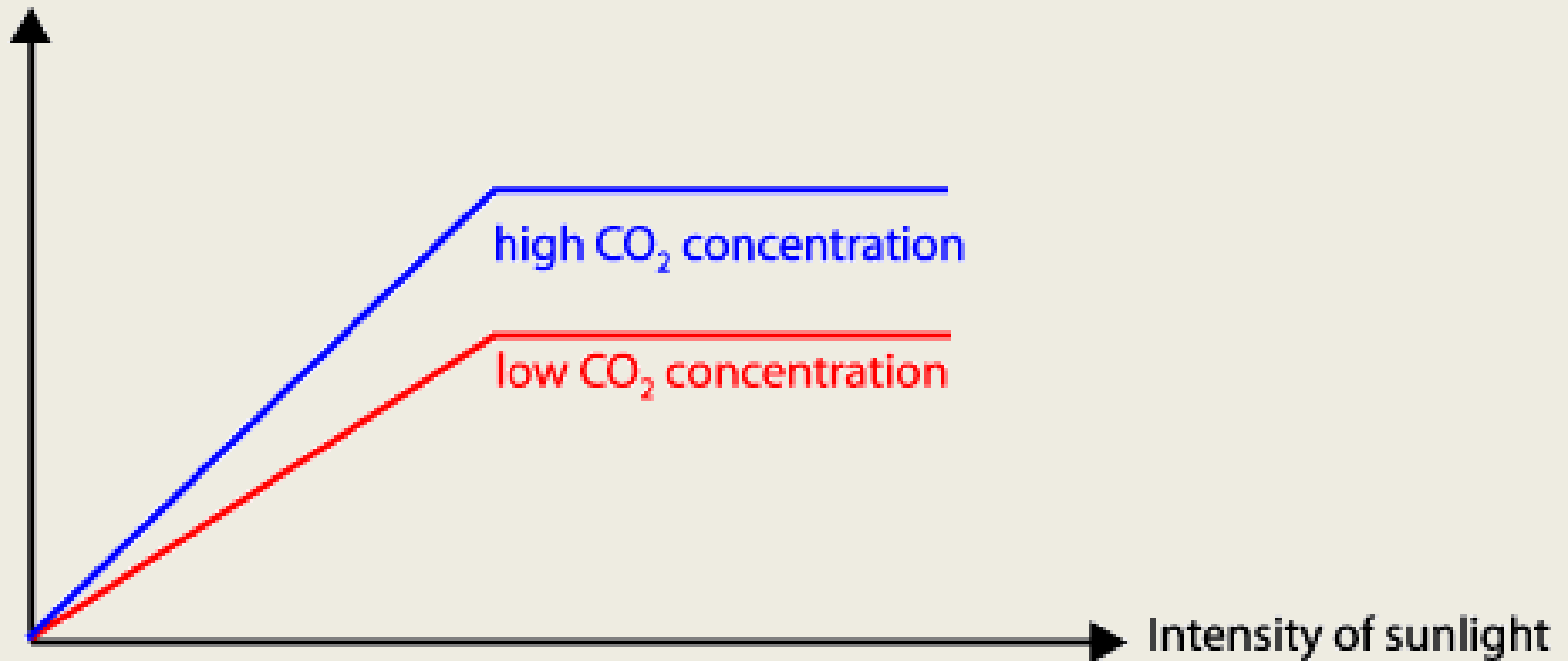


Arrange the following events in chronological order:

- A. Water molecule is split.
- B. CO_2 is converted to glucose.
- C. O_2 is produced.
- D. Light strikes the chloroplast.
- E. Energy from the light reactions runs the dark reactions.
- F. Light energy is trapped and converted.

PHOTOSYNTHESIS AND RESPIRATION

Rate of photosynthesis



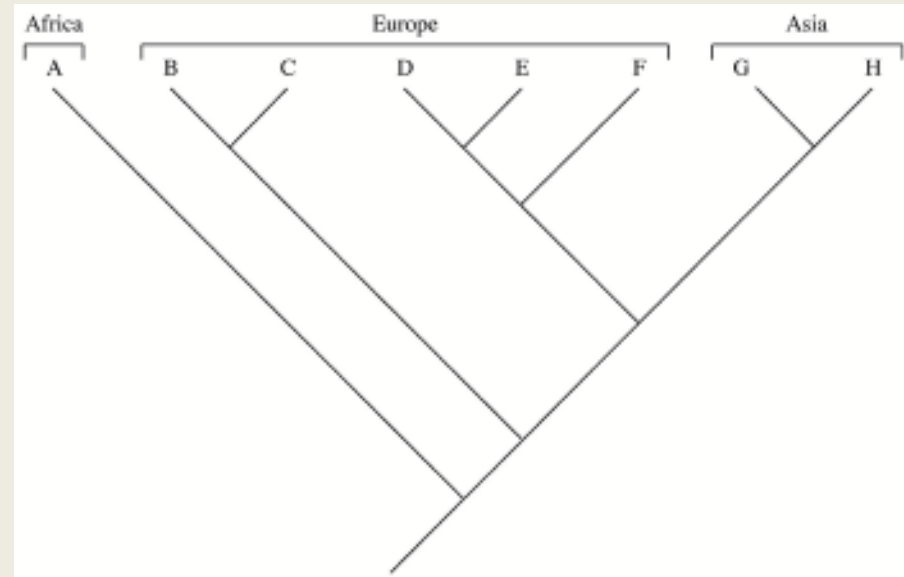
Graph two: the effect of carbon dioxide concentration on photosynthesis

- What effect does increasing sunlight intensity have on the rate of photosynthesis? Why?

EVOLUTION

■ Which statement is **CORRECT**?

- A. Species B and C evolved from species A.
- B. The genus most likely originated in Asia.
- C. Species A is the youngest taxon.
- D. Species D, E, F, G, and H share a most recent common ancestor.

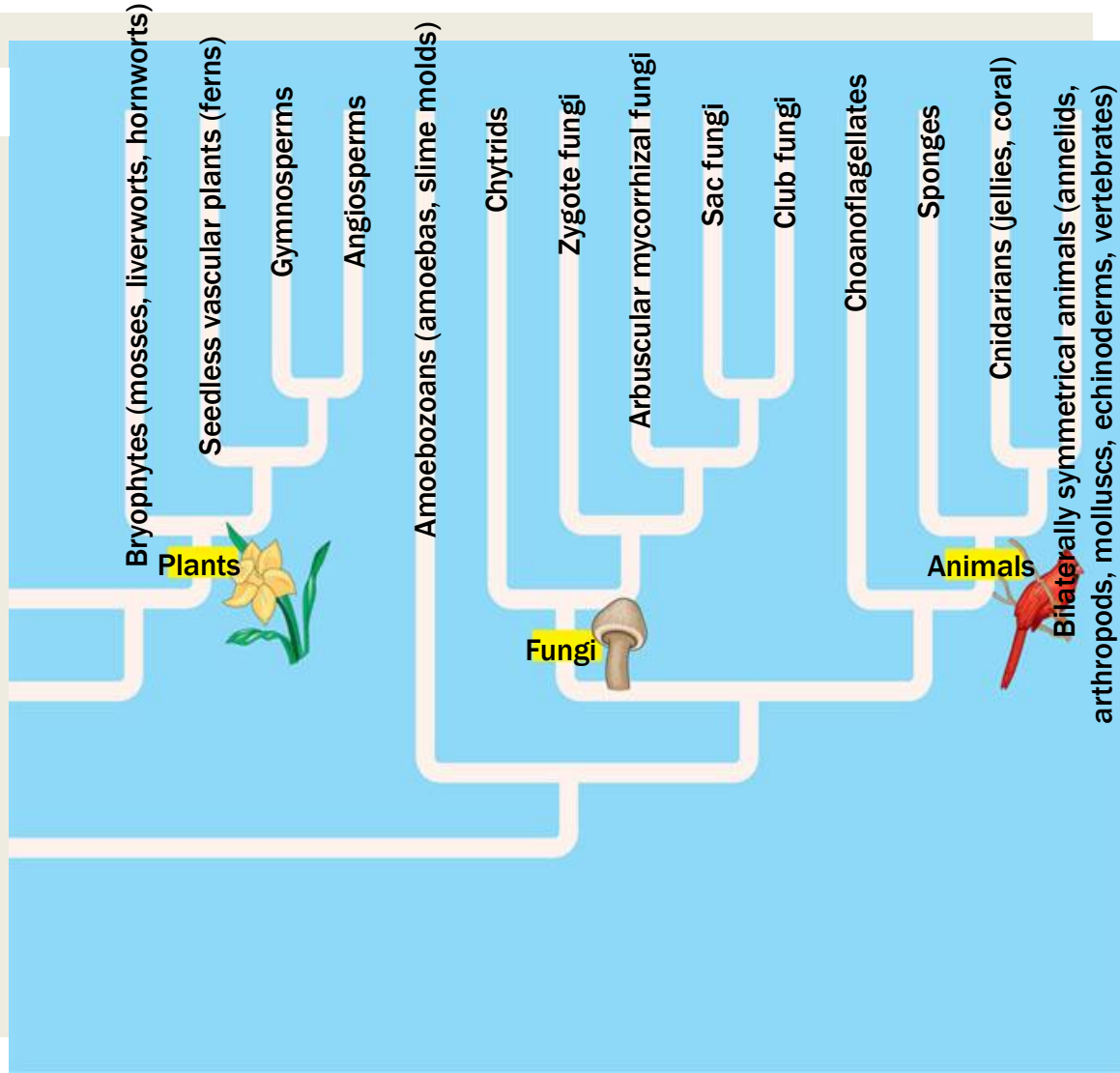


EVOLUTION

- Which domain does an organism belong to if it possesses the following characteristics?
 - Nuclear envelope - No
 - Circular chromosome - Yes
 - Flagella present - Yes
 - Cell wall material - Varies, but no peptidoglycan
 - DNA associated with histone proteins - Yes
- A. Bacteria
- B. Archaea
- C. Eukarya
- D. Bacteria or Archaea
- E. Archaea or Eukarya

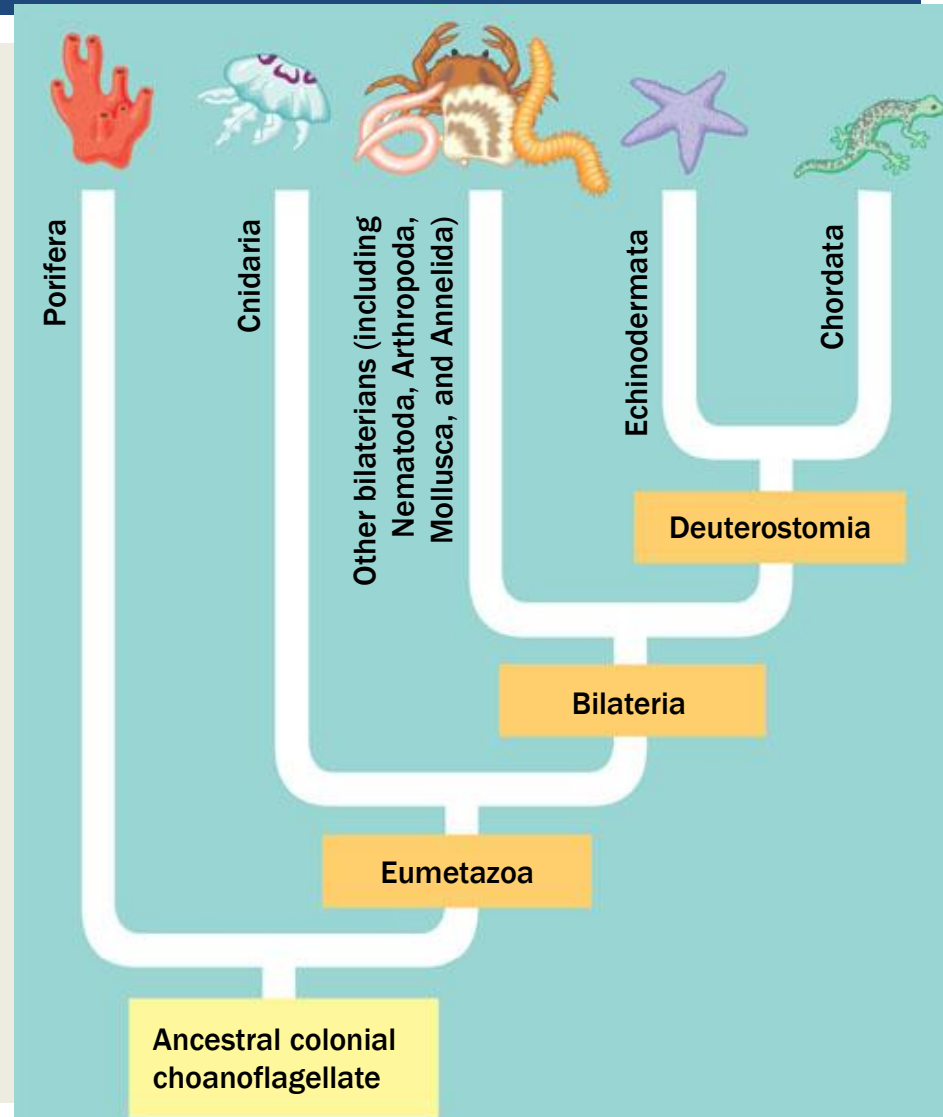
EVOLUTION

- Which characteristic is shared by fungi and animals but NOT by plants?



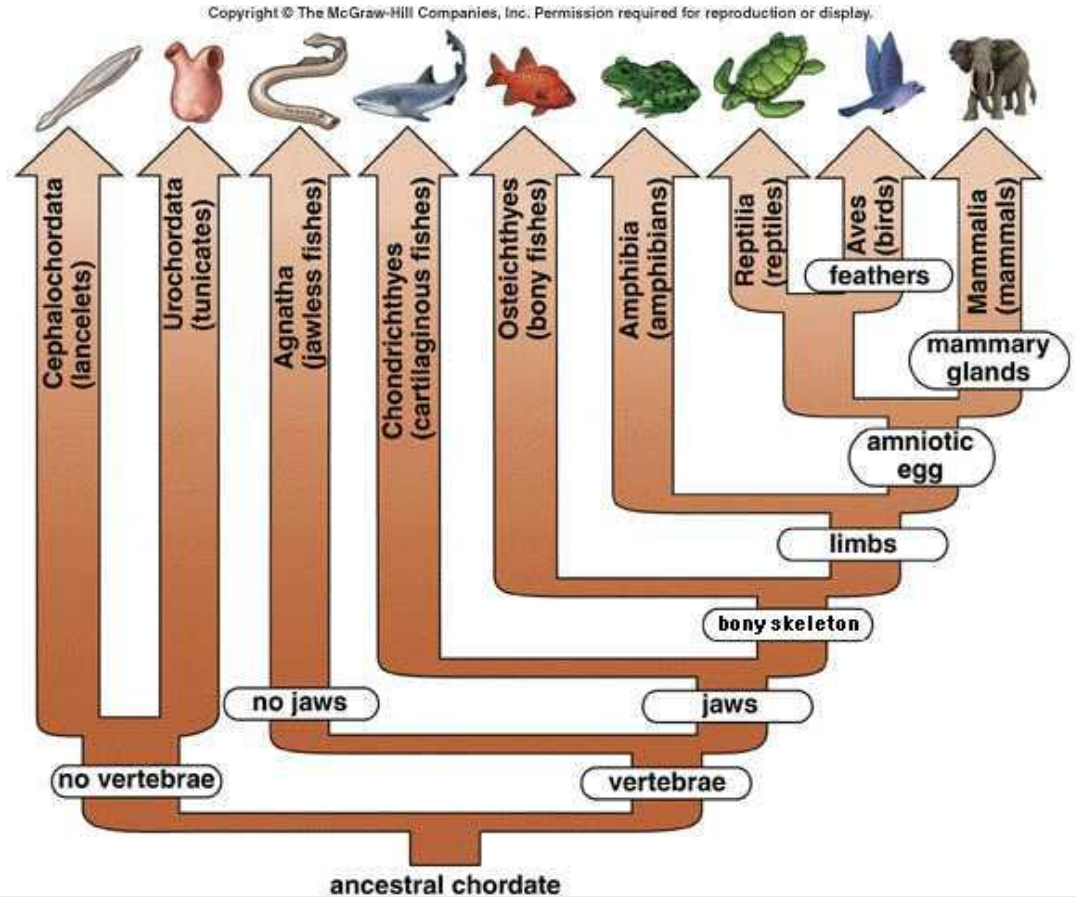
EVOLUTION

- Which characteristic is shared only by echinoderms and chordates?



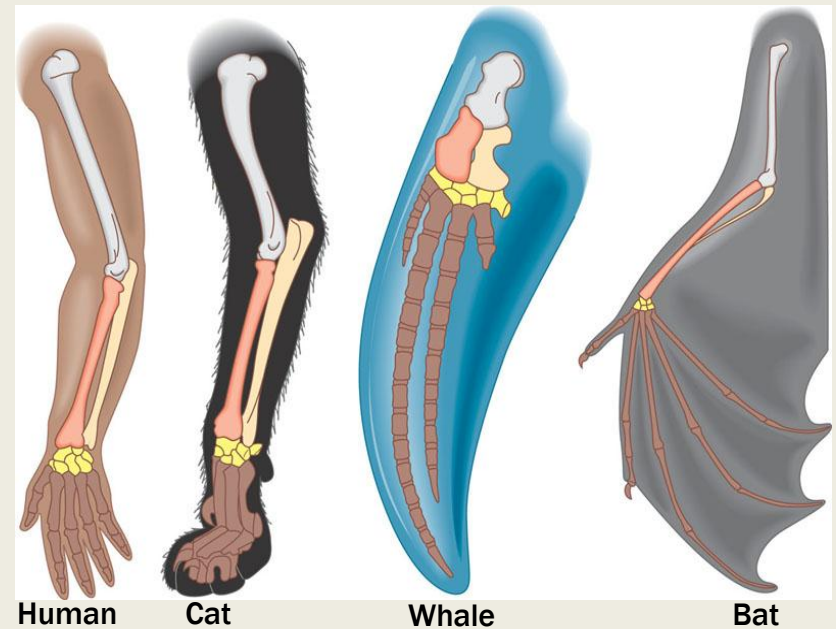
EVOLUTION

- What characteristic is shared only between reptiles, mammals, and birds?



EVOLUTION

- The structures on the right may be described as (homologous OR analogous) structures?
- What pattern of evolution may have given rise to them?



EVOLUTION

A widely-held theory is that mitochondria evolved from endosymbiotic bacteria. Indicate which statements support this theory by answering Yes (Y) or No (N).

1. Mitochondria have their own DNA
2. Mitochondria have their own ribosomes
3. Mitochondria are derived from pre-existing mitochondria by division
4. Some mitochondrial gene DNA sequences are similar to those of certain aerobic bacteria

TAXONOMY

Lemurs, chimpanzees, and humans all belong to the same order. What other, higher taxonomic levels do they have in common?

- A. Family, class, phylum
- B. Kingdom, genus, species
- C. Class, phylum, kingdom
- D. Kingdom, phylum, species
- E. Class, phylum, species

TAXONOMY

Which group of plants has vascular tissue but does not produce seeds?

I. Mosses

II. Ferns

III. Gymnosperms

A. I

B. II

C. III

D. II & III

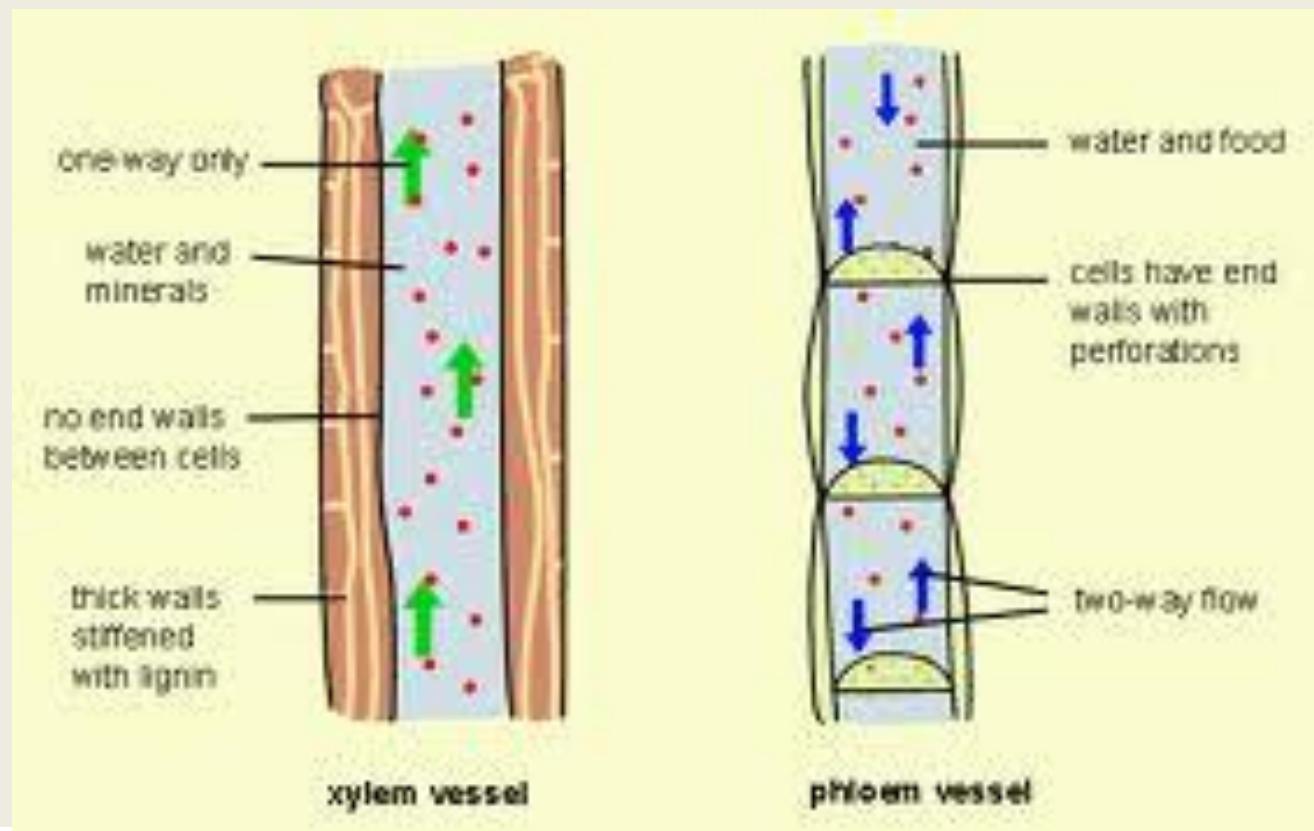
ECOLOGY

The animals at the end of a food chain are generally few in number;

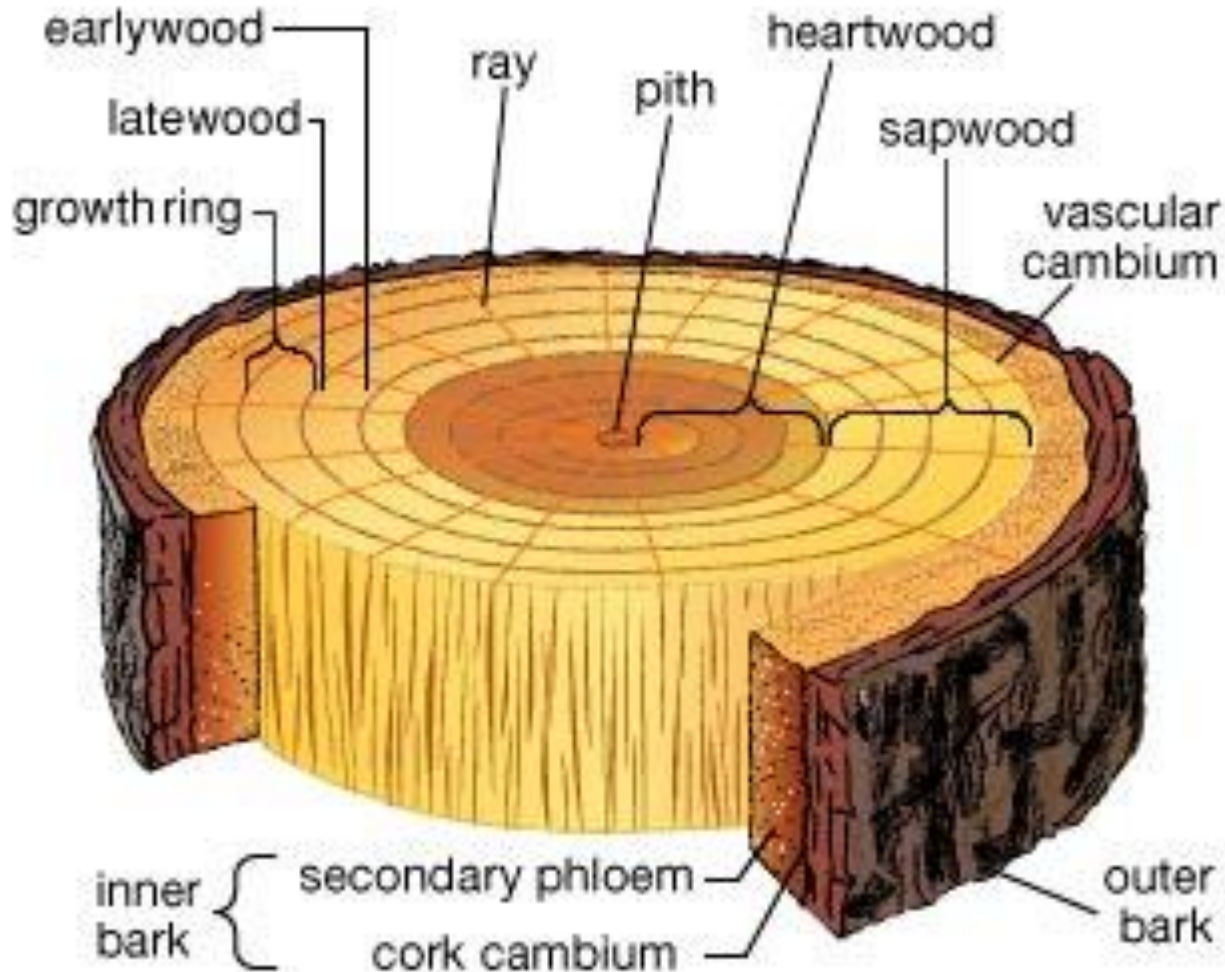
- A. Because they are always the largest organisms in the food chain.
- B. Because they have long gestation periods and few offspring.
- C. Because predators have high levels of intra-specific competition and infant mortality is high.
- D. Because of energy losses in the food chain there is insufficient energy to support large numbers of tertiary consumers.
- E. Because tertiary consumers have large territories.

CELL STRUCTURE TO FUNCTION

- How is water transport assisted by the fact that xylem vessels are dead?



CELL STRUCTURE TO FUNCTION



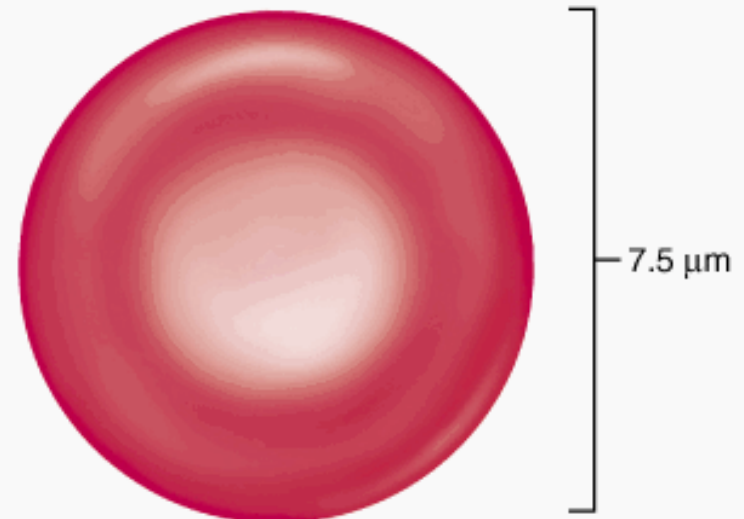
- How do stems of woody plants increase in diameter?

CELL STRUCTURE TO FUNCTION

- Why is the biconcave shape better than a spherical shape for this erythrocyte?
- What impact would the presence of a nucleus or mitochondria have on the carrying-capacity of the cell?



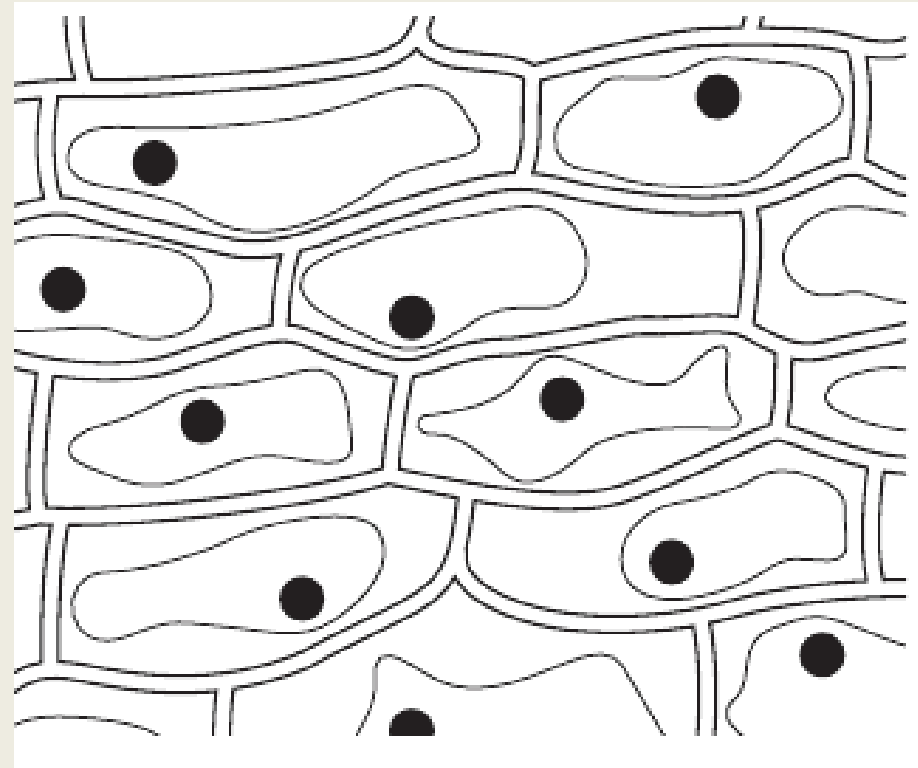
Side view



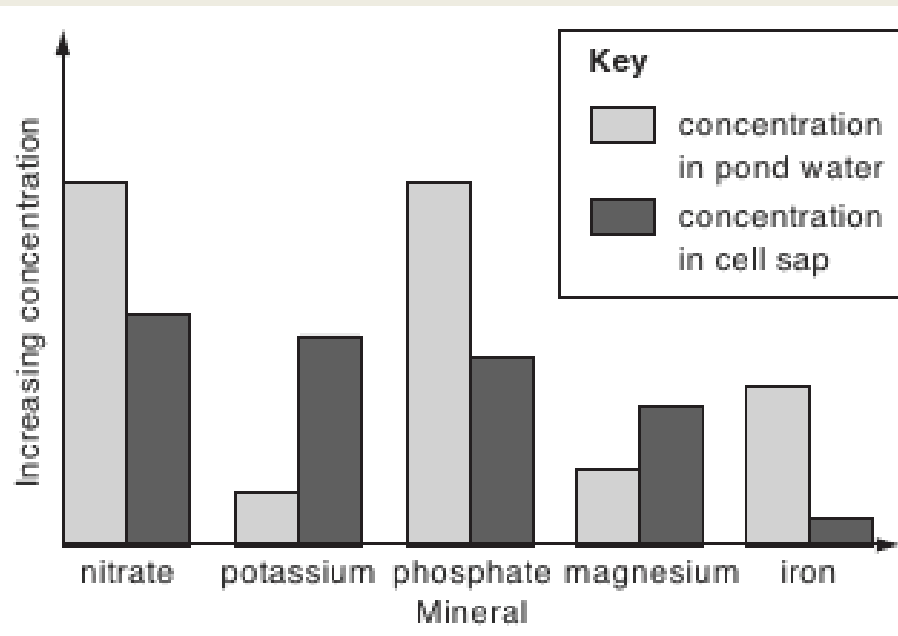
Top view

CELLULAR TRANSPORT

- The diagram shows a strip of onion tissue placed in a saturated sugar solution. What happened to cause the changes in the onion cells?



CELLULAR TRANSPORT



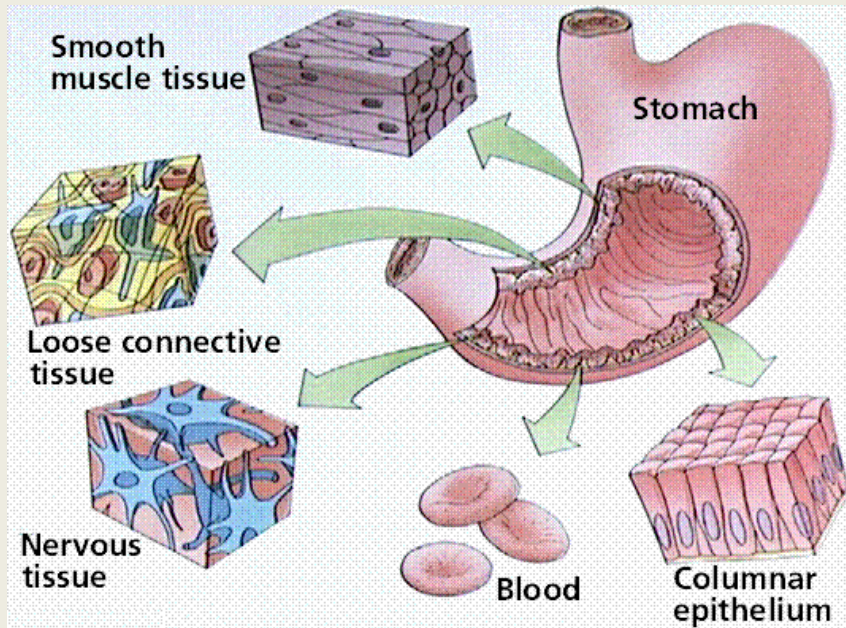
- Relative amounts of various minerals in pond water and in the cytoplasm of a microscopic green alga are shown. Which minerals must have been absorbed by active transport?

CELLULAR TRANSPORT

■ Identify mechanisms A, B, and C.

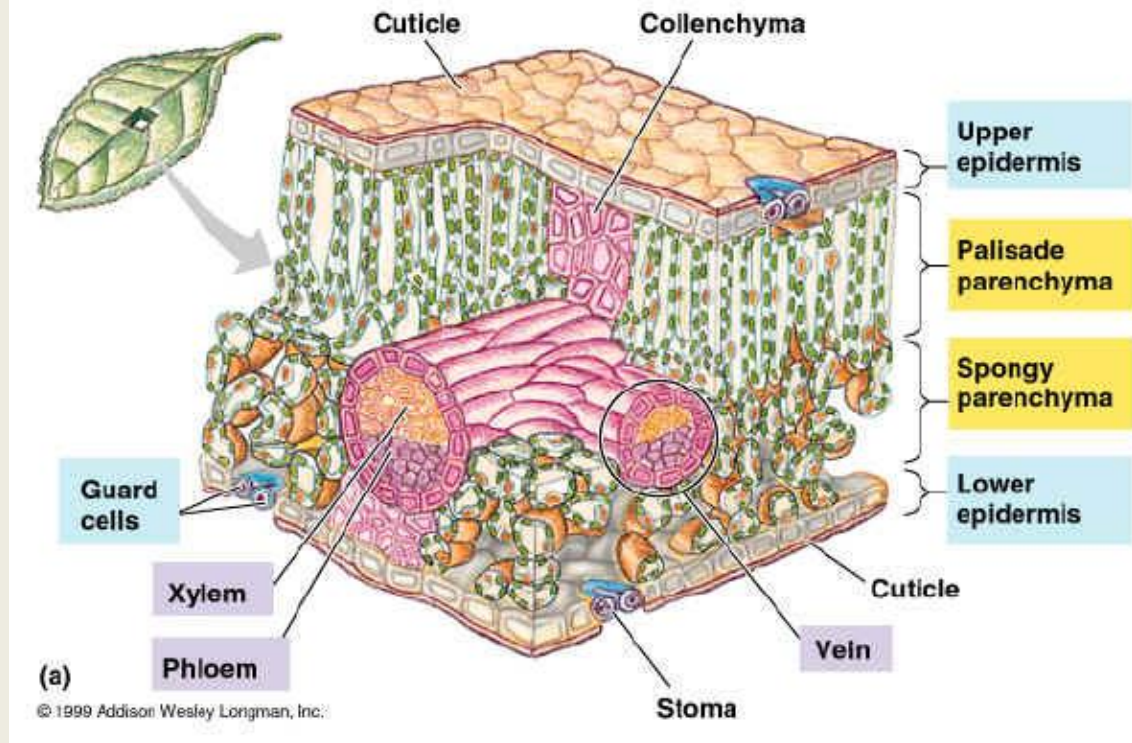
Mechanism	A	B	C
Movement down a concentration gradient	/	/	X
Takes place across a membrane	/	X	/
Requires energy input	X	X	/

TISSUE STRUCTURE TO FUNCTION



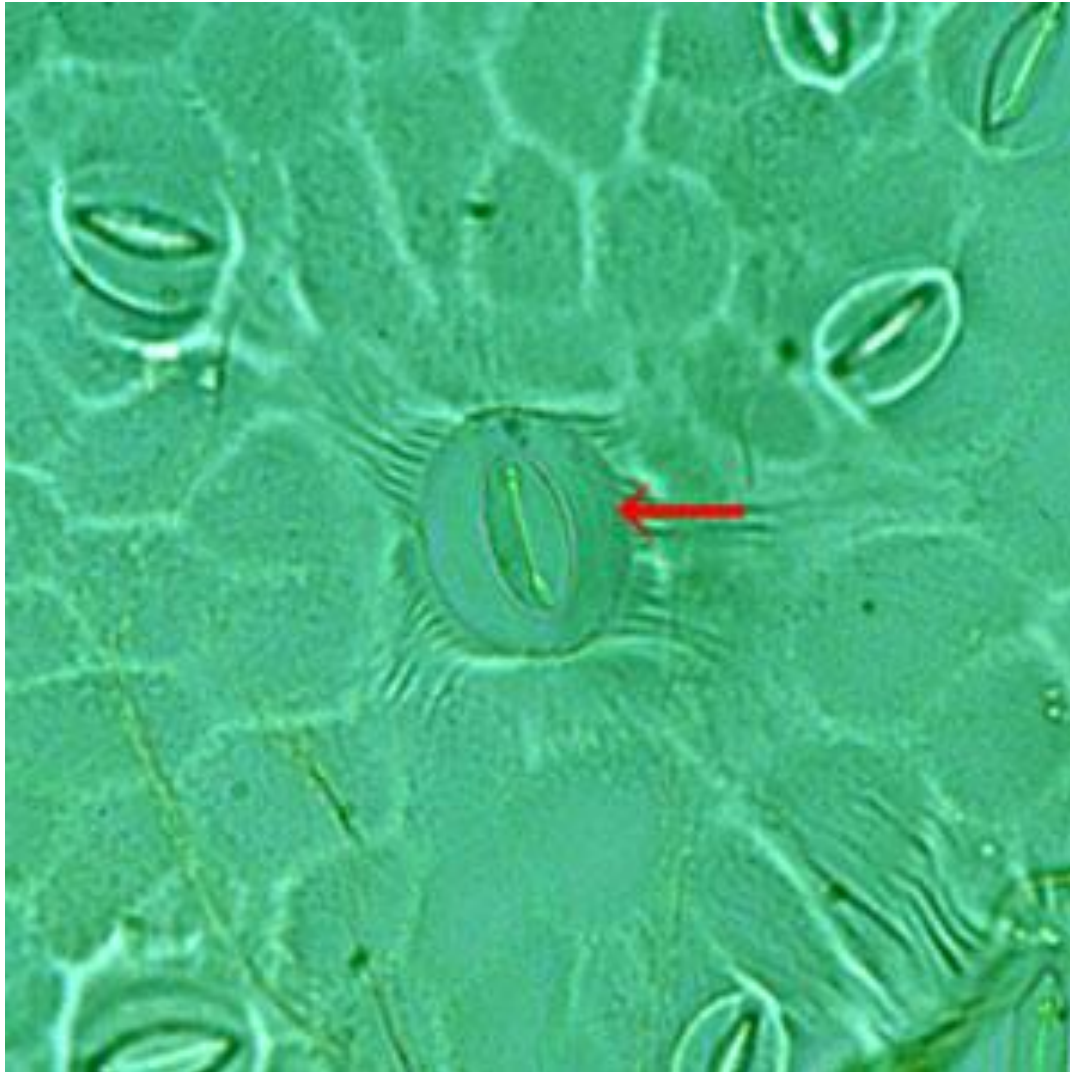
- Comes into direct contact with food, secretes gastric juice
- Carries oxygen to the cells and carries wastes away from these cells
- Activity allows food to be physically broken down into small parts
- Coordinates contraction of the stomach muscles and controls release of gastric juice

TISSUE STRUCTURE TO FUNCTION



- How do the following features of the leaf help it to photosynthesize: a) arrangement of leaves on the plant, b) thinness of leaf, c) repeated branching of veins, and d) different colors of leaves.

TISSUE STRUCTURE TO FUNCTION



- What is the function of the cells indicated by the arrow?
- A similar area on the upper surface of the leaf would have less than one-eighth of the number of these structures. Why is this an advantage?

ORGAN STRUCTURE TO FUNCTION

A heron standing in a cold water for a long time doesn't get its legs overchilled because of:

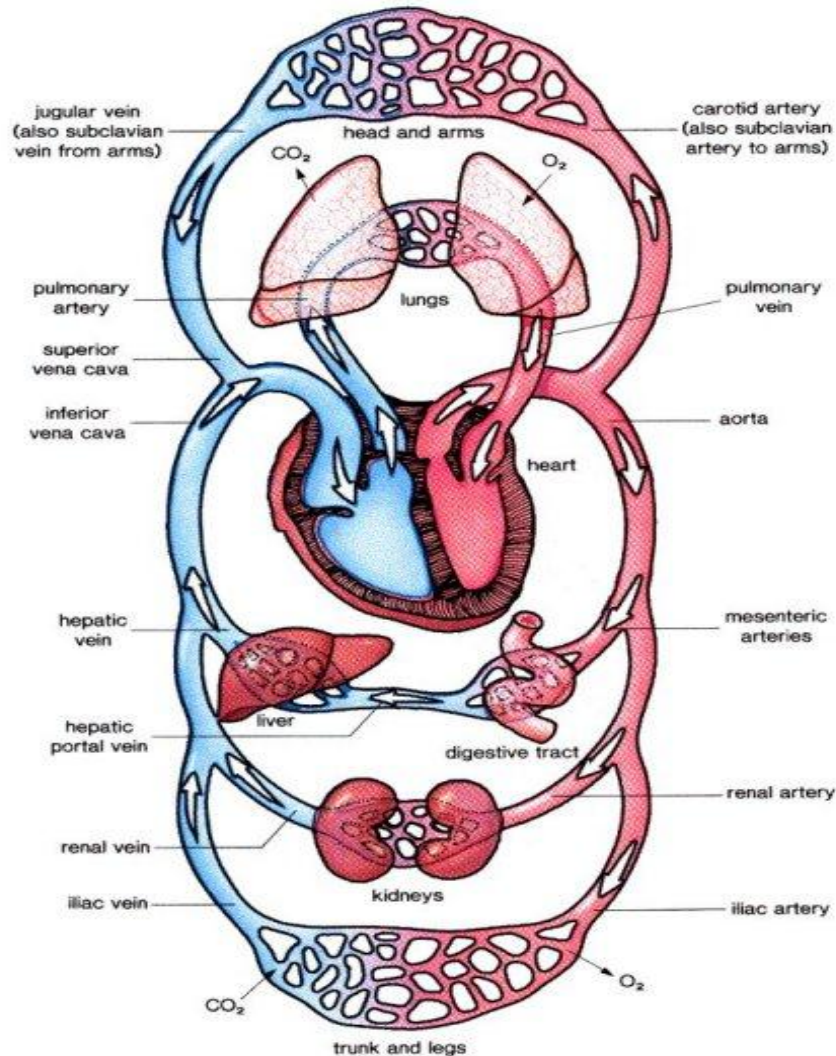
A. Increased vasodilation

B. Countercurrent circulation in limbs

C. even thin fat layer under limbs'skin

D. intensive metabolism in limbs

ORGAN STRUCTURE TO FUNCTION



Which of the following structures will not be encountered by a drop of blood going from the intestines to the brain?

- A. lung capillaries
- B. left ventricle
- C. right ventricle
- D. superior vena cava
- E. aorta